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SIGNAL: Synergistic multi-granularity semantic alignment learning for multimodal recommendation

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ABSTRACT

Multimodal recommendations play a pivotal role in online platforms by leveraging rich multimodal content to provide personalized services. To effectively exploit such content, numerous methods adopt a two-stage paradigm to learn multimodal semantics, utilizing graph neural networks for encoding and contrastive learning for alignment. However, this paradigm faces two critical challenges: (1) the neglect of synergy between encoding and alignment restricts holistic semantic learning; (2) the failure to reconcile nuanced conflicts across modalities restricts partial semantic learning. To address these challenges, we propose a **S**ynergist**I**c multi-**G**ranularity sema**N**tic **A**lignment **L**earning method (SIGNAL), which effectively learns multimodal semantics across holistic and partial granularities. Specifically, we design a pre-aligned graph encoding module that adaptively fuses modality-specific graph structures to establish preliminary alignment during encoding, and a hierarchical similarity contrastive module that preserves the encoded semantic neighborhoods during alignment. These two modules synergistically facilitate holistic semantic learning. Furthermore, we introduce a lightweight attentional diffusion module that estimates and removes semantic interference under reliability-aware guidance, thereby reconciling nuanced cross-modal conflicts. By leveraging the holistically aligned representations as a favorable initialization, the diffusion process requires only a few restoration steps, maintaining modest computational overhead. Extensive experiments demonstrate the superiority of SIGNAL over various state-of-the-art baselines on four benchmark datasets, and further verify the effectiveness of the proposed synergistic designs through visualization and robustness analysis. The code is available at https://github.com/Sail-high/SIGNAL.

1. Introduction

Recommender systems have achieved impressive performance by mining user preferences in web platforms (He & McAuley, 2016; Tao et al., 2022). As much richer item information such as texts and images becomes available, Multimodal Recommendations (MRs) have emerged as a mainstream paradigm (Liu et al., 2024; Zhou et al., 2023a). By leveraging such multimodal content, MRs can delve deeper into user preferences, thereby enhancing recommendation accuracy (Wei et al., 2023). However, raw multimodal content inevitably contains inherent noise that obscures underlying semantics, and significant semantic gaps persist among heterogeneous multimodal signals due to diverse acquisition and processing mechanisms (Cheng et al., 2019; Zhang et al., 2024). Consequently, effectively encoding multimodal semantics to mitigate content noise and aligning them with collaborative item ID embeddings remain central challenges in MRs (Xv et al., 2024).

Early feature-based methods incorporated multimodal semantics through concatenation (He & McAuley, 2016) or attention-weighted averaging (Chen et al., 2017; Liu et al., 2019), directly fusing multimodal features with item ID embeddings. However, this direct fusion of multimodal features ignores semantic representation discrepancies, and such unaligned semantics introduce noise that degrades model performance (Qi et al., 2026). Recently, structure-based methods (Zhang et al., 2021; Zhou et al., 2023b; Zhou & Shen, 2023) have shifted to constructing modality-specific item-item graph structures based on feature similarities. By aggregating information from similarity-based neighbors, Graph Neural Networks (GNNs) encode modality-specific semantics, establish corresponding semantic neighborhoods, and mitigate inherent feature

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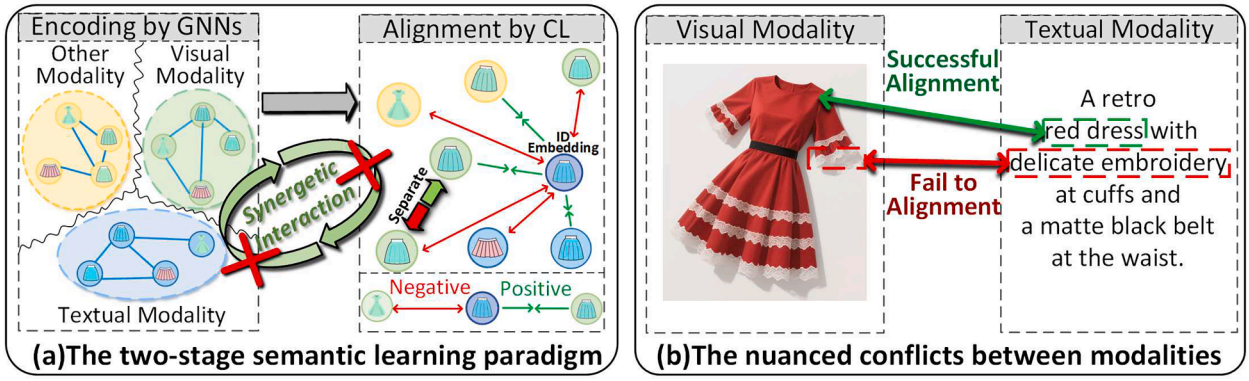


Fig. 1. (a) the two-stage paradigm, where independent GNNs encoding inherently widens semantic gaps and standard CL indiscriminately repels semantically similar items. (b) an example of nuanced conflicts: visual and textual modalities agree holistically on *red dress* but exhibit partial inconsistency, i.e., visual *lace trim* conflicting with textual *delicate embroidery*.

noise (Xv et al., 2024). To further bridge the gap between encoded multimodal semantics and collaborative item ID embeddings, extensive research integrates Contrastive Learning (CL) for alignment (Guo et al., 2024; Zhang et al., 2022). Building upon the semantics encoded by GNNs, CL aligns multimodal representations with collaborative item ID embeddings by maximizing the semantic agreement between positive pairs, while maintaining representation discriminability by treating other items as negatives and pushing them apart in the embedding space (Wei et al., 2024). Despite the progress in multimodal semantic learning driven by GNNs-based encoding and CL-based alignment, existing approaches face two primary challenges.

The neglect of synergy between encoding and alignment restricts holistic semantic learning. As illustrated in Fig. 1 (a), numerous methods typically employ GNNs to independently encode modality-specific representations; This isolation exacerbates the semantic gap across modalities, thereby creating obstacles for semantic alignment. Subsequently, during cross-modal alignment, CL uniformly treats all other items as negatives; this indiscriminate repulsion separates semantically similar items and disrupts the semantic neighborhoods established by GNNs. Therefore, synergizing semantic encoding and alignment is essential, as encoding provides a better optimization basis for subsequent alignment, while alignment preserves the semantic neighborhoods established during encoding, thereby facilitating holistic semantic learning.

The failure to reconcile nuanced conflicts across modalities restricts partial semantic learning. As illustrated in Fig. 1 (b), although both modalities holistically agree on *red dress*, a partial mismatch can be observed, where the visually observed *lace trim* diverges from the textually described *delicate embroidery*. Crucially, the coarse-grained nature of contrastive alignment primarily enforces consistency at the holistic level and thus fails to resolve such partial inconsistencies. When the modality-specific representations are subsequently fused, these conflicting semantic cues enter the item representation and obscure the semantics conveyed by each modality, resulting in semantic ambiguity. Therefore, reconciling nuanced conflicts to ensure partial consistency is essential for robust partial semantic learning.

In this paper, we propose a **S**ynergist**I**c multi-**G**ranularity sema**N**tic **A**lignment **L**earning method (SIGNAL) for MRs, which effectively learns multimodal semantics across holistic and partial granularities. Specifically, to facilitate holistic semantic learning, we design a pre-aligned graph encoding module and a hierarchical similarity contrastive module. The former introduces an inter-graph attention mechanism that adaptively fuses modality-specific graph structures, thereby producing preliminarily aligned multimodal representations during encoding. The latter stratifies item relations for differentiated contrast, thereby preserving the encoded semantic neighborhoods during alignment. Together, they establish a bidirectional encoding-alignment synergy, mitigating the optimization mismatch caused by treating the two stages as a one-way pipeline. Furthermore, to improve partial semantic learning, we design a lightweight attentional diffusion module to reconcile nuanced cross-modal conflicts. During the forward process, it constructs modality-mixed semantic noise to simulate cross-modal semantic conflicts; during the reverse process, it employs a reliability-aware guiding signal to guide conflicting features toward more reliable semantics. Besides, the holistically aligned representations obtained through contrastive learning provide a favorable initialization, allowing the diffusion process to refine partial semantic discrepancies with only a few restoration steps and modest computational overhead. The main contributions of this paper are as follows:

- We propose SIGNAL, a multi-granularity semantic alignment framework that facilitates holistic semantic learning by establishing synergy between the graph encoding and contrastive alignment stages, while improving partial semantic learning by refining nuanced cross-modal conflicts.
- For holistic semantic learning, we introduce a synergistic encoding-alignment mechanism. Inter-graph attention mechanism adaptively fuses modality-specific graph structures during encoding, while a neighborhood-guided differentiated contrastive strategy preserves the encoded semantic neighborhoods during alignment.
- For partial semantic learning, we develop a lightweight attentional diffusion mechanism that reconciles nuanced cross-modal conflicts through semantic noise simulation and reliability-aware guidance, while leveraging holistically aligned representations to reduce the iterative overhead of standard diffusion.
- Extensive experiments on four real-world datasets demonstrate the superiority of SIGNAL. Comparisons with closely related methods and ablation studies verify the effectiveness of the proposed designs, and targeted experiments further validate the synergistic mechanisms and robustness under diverse settings.

The rest of this paper is organized as follows. Section 2 reviews the related work on two types of MRs and diffusion-based MRs. Section 3 details the implementation of the proposed SIGNAL framework and provides a theoretical analysis of its synergistic advantages and time complexity. Section 4 conducts a comprehensive comparative analysis of SIGNAL against various state-of-the-art baseline methods. Finally, Section 5 concludes the paper by summarizing the contributions of SIGNAL, discussing its limitations, and outlining directions for future work.

2. Related work

This section provides a knowledge expansion and literature review concerning the model, detailing prior research in relevant areas and outlining both their contributions and limitations. The discussion is

structured into two primary parts: (1) classic multimodal recommendation, and (2) diffusion-based multimodal recommendation.

2.1. Classic multimodal recommendation

Multimodal Recommendations (MRs) aim to enhance recommendation accuracy by exploiting rich multimedia information to alleviate data sparsity (Wei et al., 2023). Consequently, a critical task for MRs lies in effective multimodal semantic learning to enrich item representations (Wei et al., 2020). From the perspective of semantic learning paradigms, existing MRs methods can be broadly categorized into two streams: feature-based methods, and structure-based methods (Xv et al., 2024).

Feature-based methods primarily focus on incorporating multimodal semantics by directly fusing raw multimodal features with item ID embeddings. Pioneering works, such as VBPR (He & McAuley, 2016) and ACF (Chen et al., 2017), employed concatenation or attention mechanisms to assimilate visual features. To further enhance the quality of semantic integration, recent approaches have adopted Contrastive Learning (CL). For instance, BM3 (Zhou et al., 2023c) leverages contrastive objectives to align multimodal features with ID embeddings, thereby ensuring better multimodal semantic consistency. Building on this, SLMRec (Tao et al., 2022) advances the paradigm by applying feature masking and multi-layer contrastive fusion, aiming to learn more robust multimodal semantics. Despite the progress achieved by these methods, the inherent noise within the modalities hinders semantic learning, thereby limiting further performance gains (Zhang et al., 2024).

Differing from feature-based methods, recent structure-based methods employ Graph Neural Networks (GNNs) to encode multimodal semantics, effectively mitigating inherent noise. LATTICE (Zhang et al., 2021) constructs item-item graph structures using feature similarities, after which it employs GNNs to aggregate neighbor information to encode multimodal semantics. Building on this, FREEDOM (Zhou & Shen, 2023) freezes the fused item-item graph, thus boosting the stability of semantic encoding and reducing computational overhead. In contrast to such unified graph encoding, DA-MRS (Xv et al., 2024) first encodes distinct multimodal semantics from modality-specific item-item graph structures, providing more flexible representations for subsequent multimodal semantic learning. Following DA-MRS, we construct modality-specific item-item graphs with cross-modal pruning and build an item co-occurrence graph from user-item interactions to incorporate behavior-aware relations. These graphs serve as the input to our framework rather than as the contributions.

To align multimodal semantics, contrastive learning has been further incorporated into structure-based methods. MICRO (Zhang et al., 2022) constructs consensus-aware self-supervised signals to align multimodal representations through CL. MGCN (Yu et al., 2023) purifies modality semantics with behavioral information and aligns them with item ID embeddings to reduce modality noise. In addition, LGMRec (Guo et al., 2024) models local and global user interests through graph and hypergraph learning, while aligning global representations across modalities through CL. NEGCL (Shi et al., 2025) introduces noise-enhanced graph contrastive learning to improve the robustness of multimodal representations. DA-MRS performs graph encoding and then selects highly similar item pairs for each modality to conduct graded contrastive learning. However, this coarse modality-wise selection ignores cross-modal consistency and can lead to divergent item representations across modalities. Building upon this idea, SIGNAL derives hierarchical item relations from the modality-specific neighborhoods established during graph encoding and their cross-modal consistency, and performs differentiated contrast to preserve the encoded semantic neighborhoods.

Despite the empirical success of structure-based methods that perform graph encoding followed by contrastive alignment, the two stages are commonly organized as a sequential pipeline with weak coupling. As a result, separate graph encoding enlarges the representation gaps across modalities, while indiscriminate contrastive repulsion disrupts the semantic neighborhoods formed during graph encoding. This limitation motivates an explicit encoding-alignment synergy in which the two stages mutually reinforce holistic semantic learning. To provide a clearer comparison from the perspective of graph encoding and contrastive alignment, Table 1 summarizes the differences between SIGNAL and representative methods. Existing methods have achieved meaningful progress in graph construction and contrastive learning, but most follow a sequential encoding-alignment pipeline and rarely establish explicit bidirectional interactions between the two stages. In contrast, SIGNAL employs inter-graph attention to promote preliminary alignment during graph encoding, providing a favorable basis for subsequent contrastive alignment. In turn, neighborhood-guided differentiated contrast preserves the semantic neighborhoods established during encoding, thereby forming bidirectional synergy between the two stages.

2.2. Diffusion models for multimodal recommendation

Diffusion Models (DMs) possess remarkable generative capabilities derived from a fundamental process of forward noise injection and reverse feature reconstruction (Ho et al., 2020). This process equips DMs with the ability to generate high-quality data, as demonstrated in image synthesis (Rombach et al., 2022) and text generation (Li et al., 2022).

In multimodal recommendation, researchers have sought to leverage DMs for data generation to enhance recommendation performance. DiffCL (Song et al., 2025) utilizes diffusion model to generate contrastive views, mitigating noise introduced by self-supervised learning. MoDiCF (Li et al., 2025a) learns modality-specific distributions to synthesize missing modality data, thereby addressing modality incompleteness and improving recommendation fairness. Recently, SDMMR (He et al., 2026) combines masked prediction with diffusion models to capture complex multimodal noise patterns, alleviating feature bias during denoising. Beyond the data augment perspective, DiffMM (Jiang et al., 2024) adopts graph diffusion to reconstruct interaction graphs, and resolves fine-grained feature discrepancies. Besides, diffusion models have also been combined with contrastive learning in knowledge-aware recommendation. KMDCL (Li et al., 2025b) adopts mask diffusion to adaptively generate knowledge views from local and global perspectives, while HDKCL (Li et al., 2026b) leverages hierarchy-aware diffusion and knowledge-enhanced contrastive learning to improve knowledge-aware recommendation.

Given the strong restoration capability of DMs, we design a lightweight diffusion-based module for mitigating nuanced cross-modal conflicts. It employs a reliability-aware guiding signal to suppress conflict-related semantic noise and leverages holistically aligned representations to reduce computational overhead. Meanwhile, recent recommendation studies have explored emerging LLM-based reasoning paradigms, such as ScotRec (Li et al., 2026c). Extending diffusion-based refinement to identify and correct inconsistencies introduced by external knowledge or intermediate reasoning processes represents a promising direction for improving the reliability of recommendation reasoning.

3. Methodology

In this section, we first formulate the problem of MRs and then present the proposed SIGNAL method. As shown in Fig. 2, SIGNAL learns multimodal semantics at holistic and partial granularities.

For holistic semantic learning, SIGNAL combines a pre-aligned graph encoding module with a hierarchical similarity contrastive module to address the weak coupling between graph encoding and contrastive alignment. The former adaptively fuses modality-specific graph structures according to cross-modal neighborhood consistency, producing preliminarily aligned representations for subsequent alignment. The latter differentiates item relations based on the semantic neighborhoods established during graph encoding, thereby aligning multimodal repre-

Table 1
Comparison of SIGNAL with representative multimodal recommendation methods.

Method	GNNs Encoding	CL Alignment and Differentiation	GNNs–CL Coupling	Partial-Conflict
MICRO	Constructs modality-specific item graphs; separately encodes them.	Aligns modality-specific representations through CL, without differentiation.	Encoding and alignment are performed sequentially, without bidirectional interaction.	No reconciliation
FREEDOM	Fuses modality-specific item graphs; uniformly encodes them.	Does not perform alignment with CL.	The framework focuses on unified graph encoding.	No reconciliation
MGCN	Constructs modality-specific item graphs; separately encodes them.	Aligns modality-specific representations through CL, without differentiation.	Encoding and alignment are performed sequentially, without bidirectional interaction.	No reconciliation
DA-MRS	Constructs modality-specific item graphs and co-occurrence graph; separately encodes them.	Aligns modality distributions via KL divergence and distinguishes item relations for graded CL.	Graded CL follows encoding, but its coarse modality-wise relation grading overlooks cross-modal consistency.	No reconciliation
LGMRec	Constructs modality-aware global hypergraphs; separately encodes them.	Aligns modality-specific representations through CL, without differentiation.	Encoding and alignment are performed sequentially, without bidirectional interaction.	No reconciliation
NEGCL	Constructs noise-enhanced modality-specific item graphs; separately encodes them.	Aligns modality-specific representations through CL, without differentiation.	Encoding and alignment are performed sequentially, without bidirectional interaction.	No reconciliation
DiffMM	Constructs modality-aware user-item graphs; separately encodes them.	Aligns modality-specific representations through CL, without differentiation.	No bidirectional GNNs-CL interaction is established.	No reconciliation
MoDiCF	Constructs modality-aware user-item graphs; separately encodes them.	Aligns modality-specific representations through CL, without differentiation.	GNNs and CL are not designed as encoding-alignment synergy.	No reconciliation
SIGNAL	Constructs modality-specific item graphs and co-occurrence graph; **interactively encodes them** through inter-graph attention to learn preliminarily aligned modality representations.	Aligns modality-specific representations through CL, with **differentiating strong and weak item relations** according to graph-encoded neighborhoods and cross-view consistency.	Encoding provides **preliminarily aligned representations** for alignment, and hierarchical contrast **preserves the semantic neighborhoods** established during encoding, **forming bidirectional encoding-alignment synergy.**	**Reconciliation** with lightweight attentional diffusion.

sentations with item ID embeddings while preserving the encoded neighborhoods.

For partial semantic learning, SIGNAL introduces a lightweight attentional diffusion module to refine nuanced cross-modal conflicts that remain after holistic alignment. Guided by reliability-aware semantics and initialized with holistically aligned representations, it focuses on partial discrepancies with only a few restoration steps and modest computational overhead.

Finally, we analyze the synergy among the proposed components and the time complexity of SIGNAL.

3.1. Problem formulation

Let $\mathcal{U}$ and $\mathcal{I}$ denote the sets of users and items, respectively. The interactions between users and items are represented by a matrix $\mathcal{O} \in \{0,1\}^{|\mathcal{U}|\times|\mathcal{I}|}$, where $\mathcal{O}_{ui} = 1$ indicates the user $u \in \mathcal{U}$ interacts with item $i \in \mathcal{I}$, otherwise $\mathcal{O}_{ui} = 0$. For each modality $m \in \mathcal{M}$, the content of item i is projected into a feature vector $\mathbf{e}_i^m \in \mathbb{R}^{d_m}$, where $\mathcal{M} = \{v, t, a\}$ denotes visual, textual, and acoustic modalities, and d_m is the embedding dimension. Given $\mathcal{O}$ and $\mathbf{e}^m$, the task of MRs is to provide a ranking list of possible recommendations that each user u may prefer, according to the predicted user-item preference score $\hat{y}_{ui}$.

3.2. Pre-aligned graph encoding

To achieve semantic pre-alignment during the semantic encoding stage, the Pre-aligned Graph Encoding (PGE) module initially builds and refines modality-specific item-item graphs. Specifically, it establishes graphs based on intra-modal feature similarity and then prunes the edges according to cross-modal information to alleviate the interference of noise. Subsequently, it fuses these graphs based on inter-graph attention to pre-align multimodal semantics of items with ID embeddings, effectively mitigating the semantic gap among various modalities.

3.2.1. Item-item graphs construction

As illustrated in Fig. 3, an item-item graph is constructed for each modality based on the similarities between modality-specific features. Specifically, for each modality $m \in \mathcal{M}$, the semantic similarity matrix $\mathbf{S}^m$ is defined as:

$$\mathbf{S}_{i,j}^m = \frac{(\mathbf{e}_i^m)^\top \mathbf{e}_j^m}{\|\mathbf{e}_i^m\| \|\mathbf{e}_j^m\|}, \tag{1}$$

where each element $\mathbf{S}_{i,j}^m$ represents the similarity between items i and j in modality m. We follow the cross-modal consistency-based pruning strategy in DA-MRS (Xv et al., 2024) to denoise the modality-specific similarity matrices. Specifically, if the similarity $\mathbf{S}_{i,j}^m$ is below the average similarity of modality m, it will be set to 0 to remove weak similarity relationships. Then, entries that are inconsistent across modalities are further discarded, so that only item pairs with relatively high similarity across all modalities are retained. This strategy mitigates the noise caused by a particular modality and constructs cleaner modality-specific item-item graphs:

$$\mathbf{S}_{i,j}^m = \begin{cases} 0, & \text{if } \exists\, m',\ \mathbf{S}_{i,j}^{m'} = 0, \\ \mathbf{S}_{i,j}^m, & \text{otherwise.} \end{cases} \tag{2}$$

Furthermore, for each item $i \in \mathcal{I}$ in modality m, the top k items most similar to it are retrieved to form a similarity list called top-k($\mathbf{S}_{i,:}^m$). And based on the semantic similarity matrix $\mathbf{S}^m$ and the list top-k($\mathbf{S}_{i,:}^m$), an adjacency matrix $\mathbf{A}^m$ for the m-modal item-item graph is defined as:

$$\mathbf{A}_{i,j}^m = \begin{cases} 1, & \mathbf{S}_{i,j}^m \in \text{top-k}(\mathbf{S}_{i,:}^m)\ \&\ \mathbf{S}_{i,j}^m > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

In addition to content-based modality graphs, the co-occurrence frequency of items clicked by the same user can also reflect important

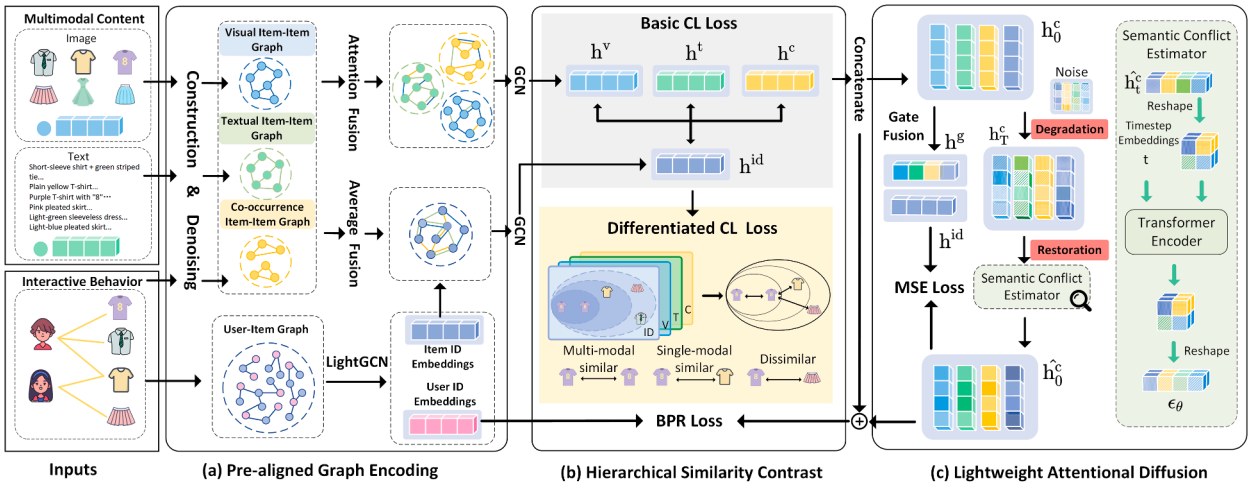


Fig. 2. The framework of the proposed SIGNAL. It consists of: (a) the PGE module, which encodes multimodal semantics while performing preliminary alignment; (b) the HSC module, which aligns holistic semantics while preserving encoded semantic structure; and (c) the LAD module, which reconciles nuanced conflicts across modalities.

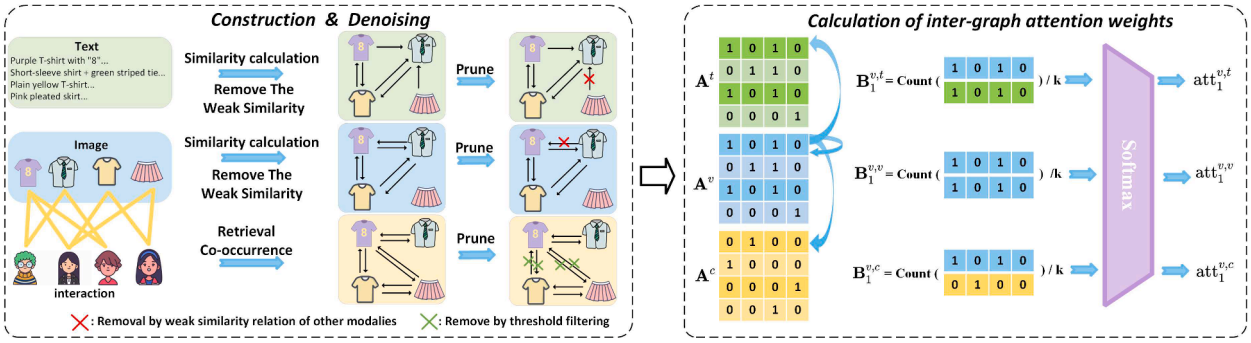


Fig. 3. Main components of the PGE module, including construction & denoising and inter-graph attention calculation. Note that the latter illustrates the attention calculation for the first row (denoted by subscript 1) only for process demonstration. In practice, the actual model computes the entire matrix.

behavior-aware item relatedness (Qi et al., 2025). Following DA-MRS (Xv et al., 2024), we construct an item-item co-occurrence graph from user-item interactions to complement content-based semantic relations. Specifically, the matrix $\mathbf{S}^c_{i,j}$ is defined as the number of shared users between items i and j. To mitigate noise from accidental clicks, a threshold ξ is applied to prune infrequent elements, i.e., $\mathbf{S}^c_{i,j} < \xi$. Then, the top-k items that have the strongest connections with item $i \in \mathcal{I}$ are retrieved to form the list top-k($\mathbf{S}^c_{i,:}$). The adjacency matrix $\mathbf{A}^c$ for the co-occurrence item-item graph is defined as:

$$\mathbf{A}^c_{i,j} = \begin{cases} 1, & (\mathbf{S}^c_{i,j} \in \text{top-k}(\mathbf{S}^c_{i,:}) \ \&\ \mathbf{S}^c_{i,j} \geq \xi) \text{ or } (i = j), \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

After constructing $\mathbf{A}^c$, we apply the same graph fusion and encoding operations to it as those applied to the content-based modality graphs. The co-occurrence graph is treated as a behavior-aware graph view rather than a conventional content modality. For notational simplicity, we augment the index set as $\mathcal{M} \leftarrow \mathcal{M} \cup \{c\}$ in the following derivations, where c denotes the co-occurrence graph.

3.2.2. Inter-graph attention fusion

To provide a more favorable optimization basis for subsequent contrastive alignment, inter-graph attention is introduced to fuse modality-specific item graphs, allowing representations from different modalities to enhance connected as they evolve through graph propagation. Intuitively, items consistently rank in top k similarity lists across multiple modalities exhibit stronger cross-modal neighborhood correlations. Building on this insight, we fuse each item-item graph with all graphs to establish multimodal semantic connections among items. Specifically, as illustrated in Fig. 3, the similarity coefficient matrix $\mathbf{B}^{m,m'}$ between modality m and each modality $m' \in \mathcal{M}$ is defined as:

$$\mathbf{B}^{m,m'} = \text{Count}(\mathbf{A}^m \ \&\ \mathbf{A}^{m'})/k, \tag{5}$$

where the function Count(&) calculate the number of shared top k neighbors of each item i between the adjacency matrices $\mathbf{A}^m$ and $\mathbf{A}^{m'}$. Normalizing these coefficients along the multimodal dimension yields inter-graphs attention weights:

$$\mathbf{att}^{m,m'} = \text{Softmax}_{m'}(\mathbf{B}^{m,:}). \tag{6}$$

These attention weights quantitatively capture the relative importance of each modality m' in reinforcing the semantic structure of the target modality m. A higher weight indicates that the graph structure of modality m' exhibits greater consistency with modality m in terms of top k neighborhood. Then, the fusion of the item-item graph structures is achieved through the adaptive weight $\mathbf{att}^{m,m'}$ and a hyperparameter α_1:

$$\mathbf{A}^m = \alpha_1 * \mathbf{A}^m + (1 - \alpha_1) * \sum\nolimits_{m' \in \mathcal{M}} \mathbf{att}^{m,m'} * \mathbf{A}^{m'}. \tag{7}$$

By incorporating neighborhood structures shared across modalities, the fused graphs enable modality-specific representations to interact throughout message propagation rather than evolve in isolation. This process preserves the original modality-specific neighborhoods while introducing complementary cross-modal cues, thereby narrowing the semantic gap before contrastive alignment and producing preliminarily aligned representations as a more coherent optimization starting point.

3.2.3. User and item representation

We adopt LightGCN (He et al., 2020) as our message-passing kernel. For multimodal features, ℓ_1 convolutional layers are stacked on the fused item-item graphs to obtain semantically pre-aligned representations of items for each modality m, i.e., $\mathbf{h}_i^t, \mathbf{h}_i^v, \mathbf{h}_i^c$. For user and item ID embeddings, they are input along with interaction records $\mathcal{O}_{ui}$ into LightGCN and encoded through ℓ_2 convolutional layers to derive user embeddings $\mathbf{h}_u$ and item ID embeddings $\mathbf{h}_i^{id}$.

To further pre-align multimodal semantics of items with ID embeddings, all item-item graph structures are fused by average pooling:

$$\mathbf{A}^p = \text{Avg}(\mathbf{A}^t, \mathbf{A}^v, \mathbf{A}^c), \tag{8}$$

where $\mathbf{A}^p$ denotes the averaged item-item adjacency matrix. Then the model stacks ℓ_1 convolutional layers on $\mathbf{A}^p$ and regards $\mathbf{h}_i^{id}$ as the initial embeddings to learn semantically pre-aligned item ID embeddings $\mathbf{h}_i^p$. Finally, a residual connection is applied to preserve proportional original information:

$$\mathbf{h}_i^{id} = \alpha_2 * \mathbf{h}_i^p + (1 - \alpha_2) * \mathbf{h}_i^{id}, \tag{9}$$

where α_2 denotes the hyperparameter of the fusion ratio.

3.3. Hierarchical similarity contrast

To align multimodal semantics while preserving the informative semantic neighborhoods established by graph encoding, the Hierarchical Similarity Contrast (HSC) first follows conventional contrastive learning to holistically align multimodal representations with item ID embeddings. Beyond conventional alignment, it hierarchically differentiates item relations according to the semantic neighborhoods established during graph encoding, thereby preserving these neighborhoods during contrastive learning.

3.3.1. Basic contrastive learning

Building on previous multimodal contrastive alignment methods (Yu et al., 2023; Zhang et al., 2022), we apply the InfoNCE loss (van den et al., 2018) to holistically align the pre-aligned multimodal representations with item ID embeddings. Item ID embeddings are regarded as anchor points, multimodal representations of corresponding items as positive samples, and multimodal representations of other items as negative samples. Given positive and negative samples, the basic contrastive loss $\mathcal{L}_b$ is calculated as:

$$\mathcal{L}_b = -\sum_{m\in\mathcal{M}} \sum_{i=1}^{|\mathcal{B}|} \log \frac{\varphi(\mathbf{h}_i^{id}, \mathbf{h}_i^m)}{\sum_{j=1}^{|\mathcal{B}|} \varphi(\mathbf{h}_i^{id}, \mathbf{h}_j^m)}, \tag{10}$$

where $\varphi(\mathbf{h}_i^{id}, \mathbf{h}_i^m) = \exp(\text{sim}(\mathbf{h}_i^{id}, \mathbf{h}_i^m)/\tau)$, where the function $\text{sim}(\mathbf{h}_i^{id}, \mathbf{h}_i^m)$ calculates the cosine similarity between ID embeddings $\mathbf{h}_i^{id}$ and multimodal embeddings $\mathbf{h}_i^m$ of items, and τ denotes the temperature hyperparameter.

3.3.2. Differentiated contrastive learning

During holistic alignment, semantically related items identified by graph encoding can be sampled as negatives and uniformly repelled, disrupting the encoded semantic neighborhoods. To address this issue, we introduce differentiated contrastive learning. Since each item directly aggregates information from its top-k neighbors, the kth nearest neighbor defines the strong similarity threshold. Items outside the top-k neighborhood but within the top-$2k$ range are regarded as weakly related because message passing allows them to retain partial semantic affinity. Accordingly, item pairs are divided into strong, weak, and ordinary relations for differentiated contrast, rather than treating all non-positive samples uniformly. To make the relation estimation more reliable and notational simplicity, item ID information is also treated as an auxiliary view for the screening process, i.e., $\mathcal{M} \leftarrow \mathcal{M} \cup \{id\}$.

Specifically, for each modality $m \in \mathcal{M}$, the similarity matrix $\mathbf{V}^m$ is calculated as illustrated in Fig. 4, where each entry $\mathbf{V}_{i,j}^m$ is defined as the cosine similarity between item i and item j, i.e., $((\mathbf{h}_i^m)^\top \mathbf{h}_j^m)/(\|\mathbf{h}_i^m\| \|\mathbf{h}_j^m\|)$. For each item i in modality m, the strong and weak similarity thresholds are defined as $\mathbf{ST}_i^m = \mathbf{V}_{i,k^{th}}^m$ and $\mathbf{WT}_i^m = \mathbf{V}_{i,2k^{th}}^m$, respectively, where k^{th} and $2k^{th}$ denote the kth and $2k$th highest similarity scores in the sorted list of item i. Then, a similarity scoring matrix $\mathbf{P}_{i,j}^m$ is constructed to record pairwise affinity metrics:

$$\mathbf{P}_{i,j}^m = \begin{cases} 2, & \mathbf{V}_{i,j}^m \geq \mathbf{ST}_i^m, \\ 1, & \mathbf{V}_{i,j}^m \geq \mathbf{WT}_i^m \ \&\ \mathbf{V}_{i,j}^m < \mathbf{ST}_i^m, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Based on the modality-specific thresholds, $\mathbf{P}_{i,j}^m$ provides a semantic relation score under modality m: a score of 2 indicates that item j belongs to the strong neighborhood of item i, a score of 1 indicates that it falls into the weak semantic neighborhood, and a score of 0 indicates that it does not meet either neighborhood threshold in this modality. Since semantic relatedness can be supported by multiple modalities, we further aggregate the modality-specific relation scores as $\mathbf{P}_{i,j} = \sum_{m\in\mathcal{M}} \mathbf{P}_{i,j}^m$ to estimate the cross-modal semantic affinity between items i and j. A larger $\mathbf{P}_{i,j}$ indicates that the relation between items i and j is supported by more modalities or stronger modality-specific neighborhoods. Accordingly, items are stratified into three groups with different semantic strengths. Specifically, we define the multimodal similar set $\mathcal{MS}_i$ if $\mathbf{P}_{i,j} \geq \rho$, where ρ is a hyperparameter, the single-modal similar set $\mathcal{SS}_i$ if $\rho > \mathbf{P}_{i,j} \geq 2$, and the dissimilar set $\mathcal{DS}_i$ if $\mathbf{P}_{i,j} < 2$. Here, $\mathcal{MS}_i$ contains items with stronger cross-modal semantic support, $\mathcal{SS}_i$ contains items that are semantically related but supported by weaker or fewer modality-specific relations, and $\mathcal{DS}_i$ contains ordinary negative samples. According to the differentiated similarity among items, the differentiated contrastive loss is designed as:

$$\mathcal{L}_d = -\sum_{m\in\mathcal{M}} \sum_{i=1}^{|\mathcal{B}|} \Big(\log \frac{\sum_{j\in\mathcal{MS}_i} \varphi(\mathbf{h}_i^m, \mathbf{h}_j^m)}{\sum_{k=1}^{|\mathcal{B}|} \varphi(\mathbf{h}_i^m, \mathbf{h}_k^m)} + \log \frac{\sum_{j\in\mathcal{SS}_i} \varphi(\mathbf{h}_i^m, \mathbf{h}_j^m)}{\sum_{k\in\mathcal{SS}_i} \varphi(\mathbf{h}_i^m, \mathbf{h}_k^m) + \sum_{t\in\mathcal{DS}_i} \varphi(\mathbf{h}_i^m, \mathbf{h}_t^m)} \Big). \tag{12}$$

Finally, the contribution magnitude of the contrastive losses is regulated by hyperparameters λ_{cl}^b and λ_{cl}^d, which balances the alignment and differentiation of semantics:

$$\mathcal{L}_{cl} = \lambda_{cl}^b * \mathcal{L}_b + \lambda_{cl}^d * \mathcal{L}_d. \tag{13}$$

Overall, PGE and HSC establish bidirectional synergy between graph encoding and contrastive alignment, enabling coherent holistic semantic learning through preliminary alignment and neighborhood preservation.

3.4. Lightweight attentional diffusion

Although PGE and HSC facilitate holistic semantic alignment, nuanced cross-modal conflicts can remain at the partial granularity. For example, visual and textual modalities can agree on a *red dress* while describing a local attribute inconsistently. When such conflicting cues are integrated into a shared item representation, one modality can interfere with the semantics conveyed by another. To address this issue, we propose a Lightweight Attentional Diffusion (LAD) module. LAD models conflict-induced semantic interference as noise and restores conflicting features under reliability-aware guidance. Modality-mixed semantic noise exposes cross-modal conflicts, while residual refinement preserves useful modality-specific information. Initialized with holistically aligned representations, LAD requires only a few restoration steps to refine the remaining discrepancies.

3.4.1. Core operations of LAD

Standard diffusion models typically employ content-independent Gaussian noise for forward corruption. In contrast, cross-modal semantic interference arises from inconsistent or disproportionately emphasized

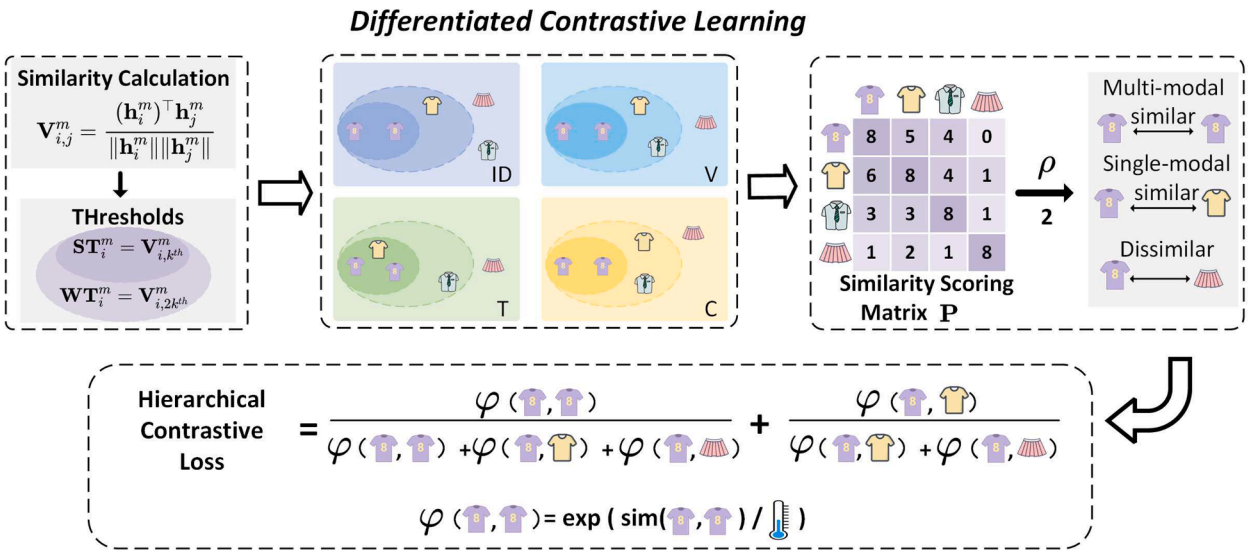


Fig. 4. Details of the differentiated contrastive learning.

modality cues of the same item, which compete when integrated into a shared representation. Therefore, directly injecting standard Gaussian noise, which is independent of item content and modality semantics, is less suitable for simulating such semantic conflicts. Cold Diffusion (Bansal et al., 2023) generalizes the diffusion process beyond the standard content-independent Gaussian corruption through a degradation operator $\mathcal{D}$ and a restoration operator $\mathcal{R}$. This formulation is suitable for LAD because it allows the degradation process to model content-dependent semantic interference. Accordingly, we construct modality-mixed semantic noise from the modality-specific representations of each item:

$$\boldsymbol{\epsilon}_i = \sum\nolimits_{m\in\mathcal{M}} \epsilon_i^m \mathbf{h}_i^m, \quad \epsilon_i^m \sim \mathcal{N}(0,1), \tag{14}$$

where ϵ_i^m is a Gaussian coefficient sampled for item i under modality m and shared across embedding dimensions. Unlike element-wise Gaussian noise, each coefficient controls an entire modality representation, allowing the perturbation to retain coherent modality-level semantics. Varying these coefficients simulates different interference patterns by changing the relative contributions of modalities.

Subsequently, the **forward diffusion process** simulates different degrees of cross-modal semantic interference by injecting the modality-mixed noise $\boldsymbol{\epsilon}_i$ over timesteps $t \in \{1, 2, \ldots, T\}$, progressively degrading the original representation $\mathbf{h}_{i,0}^m$ into $\mathbf{h}_{i,t}^m$. This degradation process generates corrupted representations with varying intensities of modality-induced semantic interference, enabling the restoration model to learn to identify and suppress semantic components that deviate from the cross-modal consensus. Since the processing is the same for each item $i \in \mathcal{I}$, we omit the index i, and simplify the notation $\mathbf{h}_{i,t}^m$ to $\mathbf{h}_t^m$ and the notation $\boldsymbol{\epsilon}_i$ to $\boldsymbol{\epsilon}$, for clarity. The degradation operator $\mathcal{D}$ formulates this process as:

$$\mathbf{h}_t^m = \mathcal{D}(\mathbf{h}_0^m, t) = \sqrt{\bar{\gamma}_t}\ \mathbf{h}_0^m + \sqrt{1-\bar{\gamma}_t}\ \boldsymbol{\epsilon}, \tag{15}$$

where the parameter $\bar{\gamma}_t$ (Jiang et al., 2024) controls the degradation operator $\mathcal{D}$ and regulates the infusion of semantically hybridized noise $\boldsymbol{\epsilon}$ and original item embeddings $\mathbf{h}_0^m$. To regulate the amount of noise injected, a linear noise scheduling mechanism is implemented for $1-\bar{\gamma}_t$:

$$1-\bar{\gamma}_t = s[\gamma_{min} + \frac{t-1}{T-1}(\gamma_{max} - \gamma_{min})], \tag{16}$$

where $s \in [0, 1]$ controls the noise scale, $\gamma_{\max}$ and $\gamma_{\min}$ represent the maximum and minimum limits of additive noise.

During the **reverse diffusion process**, the model predicts the modality-mixed semantic noise $\boldsymbol{\epsilon}_\theta$ through the Semantic Conflict Estimator (SCE) and removes it to restore refined embeddings. The restoration operator $\mathcal{R}$ formulates this process as:

$$\hat{\mathbf{h}}_0^m = \mathcal{R}(\mathbf{h}_t^m, t) = \frac{\mathbf{h}_t^m - \sqrt{1-\bar{\gamma}_t}\ \boldsymbol{\epsilon}_\theta}{\sqrt{\bar{\gamma}_t}}. \tag{17}$$

In the next subsection, we introduce the SCE and explain in detail how the degradation and restoration operators $\mathcal{D}$ and $\mathcal{R}$ are used to mitigate nuanced conflicts across modalities.

3.4.2. Semantic conflict estimator

Mitigating nuanced cross-modal conflicts requires identifying relatively reliable modality features and guiding conflicting features toward them. Accordingly, we employ a lightweight Transformer-based SCE (Vaswani et al., 2017) to estimate the modality-mixed semantic noise $\boldsymbol{\epsilon}_\theta$ through two parallel paths. The first adaptively fuses modality-specific representations with a gating mechanism to construct a reliability-aware guiding signal. The second concatenates and corrupts the item representations, which are then fed into SCE for noise estimation. The restoration operator removes the estimated noise to recover the representations. The restored ID representation is constrained by the original ID embedding, while the restored multimodal representations are guided by the reliability-aware signal. The noise-prediction objective trains SCE to estimate the injected noise, whereas the guidance objective encourages conflict mitigation toward relatively reliable semantics.

To realize this process, we first generate a reliability-aware signal $\mathbf{h}^g$ via gated fusion to guide the network in identifying the relatively reliable modality:

$$\mathbf{h}^g = \text{GateFuse}(\{\mathbf{h}_0^{m_j}\}_{j=1}^{|\mathcal{M}|}). \tag{18}$$

The reliability is implicitly learned from recommendation supervision. Since the modality-specific representations are optimized to model observed user–item interactions, modalities that provide semantics more consistent with collaborative preference signals contribute more effectively to ranking interacted items. Accordingly, the gating mechanism assigns larger weights to these relatively informative modalities and suppresses noisy or conflicting ones, making $\mathbf{h}^g$ a reliability-aware reference for subsequent restoration.

Simultaneously, we concatenate the embeddings of all modalities to form $\mathbf{h}_0^c \in \mathbb{R}^{|\mathcal{B}|\times|\mathcal{M}|\times d}$ and inject semantically hybridized noise via degradation operator $\mathcal{D}$ to yield the corrupted state $\mathbf{h}_t^c$ as formulated in Eq. (15). Next, $\mathbf{h}_t^c$ and the sinusoidal positional encoding of t are fed into fed into the SCE. Internally, the attention mechanism models

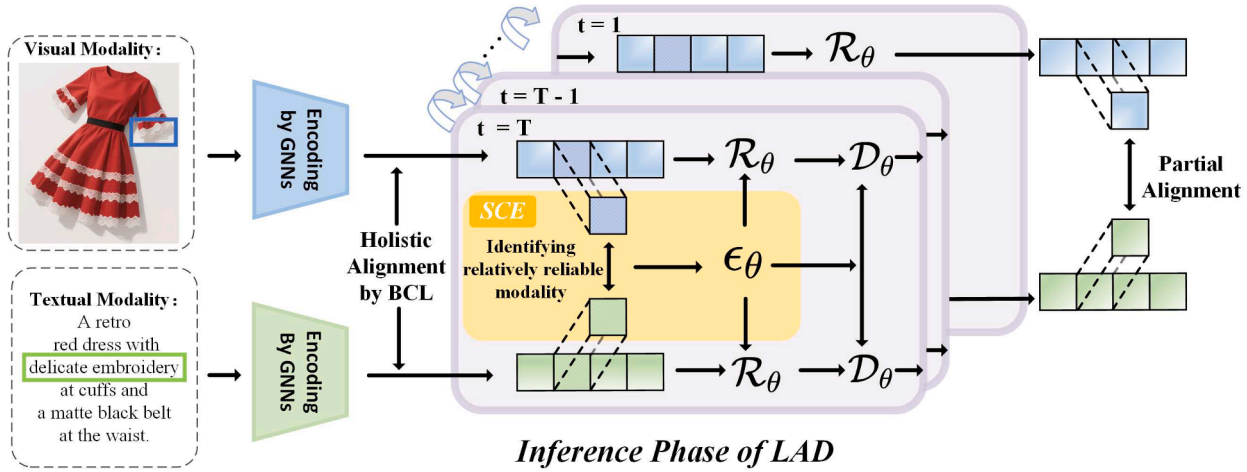


Fig. 5. Details of the Inference Phase of LAD. For clarity, only the visual and textual modalities are depicted. In the actual implementation, the model concatenates and inputs all modalities simultaneously.

cross-modal correlations to selectively aggregate contextual information. Under the supervision of the reliability-aware guiding representation, SCE is trained to emphasize semantic components consistent with relatively reliable modalities and identify deviations in the corrupted input, thereby estimating the conflict-related noise:

$$\boldsymbol{\epsilon}_\theta = \mathrm{SCE}(\mathbf{h}_t^c, t). \tag{19}$$

Subsequently, the restoration operator $\mathcal{R}$ removes this estimated noise to reveal the aligned features $\hat{\mathbf{h}}_0^c$, as formulated in Eq. (17).

Finally, the restored representation $\hat{\mathbf{h}}_0^c$ is partitioned into the restored ID representation $\hat{\mathbf{h}}_0^{id}$ and the restored non-ID multimodal representations $\hat{\mathbf{h}}_0^m$, based on which two complementary objectives are formulated:

$$\mathcal{L}_{dm_1} = E_{\mathbf{h}_0^{id}, \mathbf{h}_t^{id}}[||\mathbf{h}_0^{id} - \hat{\mathbf{h}}_0^{id}||^2]. \tag{20}$$

$$\mathcal{L}_{dm_2} = E_{\mathbf{h}_0^{m}, \mathbf{h}_t^{m}}[||\mathbf{h}^{g} - \hat{\mathbf{h}}_0^{m}||^2]. \tag{21}$$

We optimize the noise prediction capability to recover original features by minimizing the MSE between original item ID embeddings $\mathbf{h}_0^{id}$ and diffused embeddings $\hat{\mathbf{h}}_0^{id}$ via Eq. (20). Concurrently, we optimize the partial alignment process by minimizing Eq. (21). Here, the gating signal $\mathbf{h}^g$ serves as a supervisor, guiding the network to identify reliable modalities and direct the diffusion of conflicting features towards them for alignment. Although Eq. (21) encourages the restored modality representations toward a shared reliability-aware target, it does not directly replace the original modality representations. The subsequent residual refinement retains modality-specific information and controls the alignment strength.

3.4.3. Inference phase of LAD

To accelerate inference of LAD and enable partial-granularity semantic alignment, we synergistically integrate the diffusion process with the contrastive alignment. This efficiency is realized by leveraging the phased nature of the reverse diffusion process: the early steps establish the holistically coarse feature structure, while the later steps refine partial details (Kim et al., 2023; Song et al., 2021). Since basic contrastive learning (introduced in Section 3.3.1) has already established a holistically aligned semantic structure, its output provides LAD with a coarse semantic initialization. LAD can therefore focus on refining the remaining nuanced cross-modal discrepancies using only a few restoration steps, thereby maintaining modest computational overhead.

Specifically, during the inference phase of LAD, due to inherent cross-modal nuanced conflicts present in the holistically aligned embeddings $\mathbf{h}_0^c$, we do not perform explicit noise injection. Instead, we treat the embedding as if it already contains noise at timestep T, and directly proceed with iterative restoration starting from $\mathbf{h}_T^c$. Following Cold Diffusion (Bansal et al., 2023), we iteratively apply the restoration operator $\mathcal{R}$ from step $t = T$ down to $t = 1$ as illustrated in Fig. 5. The intermediate restoration process can be formulated as:

$$\hat{\mathbf{h}}_{t-1}^c = \mathcal{D}_\theta(\mathcal{R}_\theta(\hat{\mathbf{h}}_t^c, t), t-1). \tag{22}$$

The iterative process applies the restoration operator $\mathcal{R}_\theta$ at timestep t to predict and remove noise $\boldsymbol{\epsilon}_\theta$ from $\hat{\mathbf{h}}_t^c$, followed by the degradation operator $\mathcal{D}_\theta$ injecting this predicted noise $\boldsymbol{\epsilon}_\theta$ to yield $\hat{\mathbf{h}}_{t-1}^c$ at timestep $t-1$. This cycle continues until $t = 1$, culminating in $\hat{\mathbf{h}}_0^c = \mathcal{R}_\theta(\hat{\mathbf{h}}_1^c, 1)$. Through this mechanism, we achieve partial-granularity semantic alignment of multimodal embeddings.

Pure reconstruction objectives can over-smooth representations toward an average solution, potentially weakening useful modality-specific information (Mathieu et al., 2015; Zhang et al., 2017). This issue is particularly important in multimodal recommendation, because modality-specific features, such as visual style or textual attributes, still provide useful signals for distinguishing user preferences. To avoid this issue, we decompose $\mathbf{h}_0^c$ and $\hat{\mathbf{h}}_0^c$ into modality-specific features $\mathbf{h}_0^m$ and $\hat{\mathbf{h}}_0^m$, respectively, and then implement residual connections:

$$\mathbf{h}_i^m = \mu * \hat{\mathbf{h}}_{i,0}^m + (1-\mu) * \mathbf{h}_{i,0}^m, \tag{23}$$

where μ is a control hyperparameter, and $\hat{\mathbf{h}}_{i,0}^m$ and $\mathbf{h}_{i,0}^m$ are the complete representations of $\hat{\mathbf{h}}_0^m$ and $\mathbf{h}_0^m$ with index i, respectively. The residual coefficient μ controls the refinement strength. A smaller μ preserves more original modality-specific information, while a larger μ introduces stronger diffusion-based refinement. Thus, LAD conducts controlled partial refinement rather than aggressive modality homogenization, reducing the risk of over-smoothing useful modality-specific semantics.

3.5. Optimization objectives

The final item representation is obtained by averaging the ID and multimodal embeddings, i.e, $\mathbf{h}_i$=Avg($\mathbf{h}_i^{id}$, Avg($\mathbf{h}_i^t, \mathbf{h}_i^v, \mathbf{h}_i^c$)). For optimization, the Bayesian Personalized Ranking (BPR) loss (Rendle et al., 2009) is applied, based on the assumption that observed pairs should be ranked higher than unobserved ones. The BPR function is calculated as:

$$\mathcal{L}_{bpr} = -\sum_{(u,i,j)\in\mathcal{O}} \log\sigma(\mathbf{h}_u^T\mathbf{h}_i - \mathbf{h}_u^T\mathbf{h}_j), \tag{24}$$

where (u, i) denotes observed user-item pair, (u, j) denotes unobserved pair, and σ denotes the sigmoid function. The overall optimization objective is formally expressed as:

$$\mathcal{L} = \mathcal{L}_{bpr} + \mathcal{L}_{cl} + \lambda_{dm}(\mathcal{L}_{dm_1} + \mathcal{L}_{dm_2}) + \lambda_r||\Theta||^2, \tag{25}$$

Table 2
Summary of the bidirectional synergies in SIGNAL.

Synergistic Direction	Synergistic Advantages
Encoding ↔ Alignment	
PGE → HSC	Inter-graph attention produces preliminarily aligned representations, providing a favorable basis for subsequent contrastive alignment.
HSC → PGE	Graph-neighborhood-guided differentiated contrast preserves the semantic neighborhoods established during graph encoding.
Holistic ↔ Partial	
CL → LAD	Holistically aligned representations provide a favorable initialization, enabling efficient conflict refinement with only a few restoration steps.
LAD → CL	Reliability-guided diffusion mitigates nuanced cross-modal conflicts, complementing the partial semantic limitations of holistic alignment.

where λ_r and Θ represent the regularization coefficient and model parameters, respectively.

3.6. Theoretical synergy and complexity analysis

In this section, we provide a theoretical analysis of the synergistic advantages within SIGNAL and discuss their computational efficiency.

3.6.1. Analysis of synergistic advantages

Table 2 summarizes the two bidirectional synergies in SIGNAL: encoding–alignment synergy for holistic semantic learning and holistic–partial synergy for conflict refinement. Together, they connect the proposed modules across learning stages and semantic granularities. Their effectiveness is further examined through the subsequent ablation and targeted analyses.

3.6.2. Analysis of time complexity

We analyze the theoretical time complexity of SIGNAL by examining its three major components, i.e., the PGE module, the HSC module, and the LAD module.

In PGE, item–item graphs are constructed and fused once during preprocessing. This process includes multimodal similarity computation, cross-modal graph denoising, top-k graph construction, co-occurrence graph construction, and inter-graph fusion. For multimodal similarity matrix computation, SIGNAL computes an item-item similarity matrix with a complexity of $O(|\mathcal{M}||\mathcal{I}|^2 d)$, where $|\mathcal{M}|$ is the number of modalities, $|\mathcal{I}|$ is the number of items, and d is the embedding dimension. The time complexity of cross-modal graph denoising on dense similarity matrices is $O(|\mathcal{M}|^2|\mathcal{I}|^2)$. For top-$k$ graph construction, each item selects its top-k neighbors from the corresponding similarity matrix, leading to a complexity of $O(|\mathcal{M}||\mathcal{I}|\mathrm{TopK}(|\mathcal{I}|,k))$, which can be approximated as $O(|\mathcal{M}||\mathcal{I}|^2 \log k)$ when using a top-k selection routine. The complexity of constructing the co-occurrence graph is $O(|\mathcal{I}|^2|\bar{E}_i|)$, where $|\bar{E}_i|$ is the average number of user interactions per item, since the implementation computes item-item co-occurrence by checking user-set intersections for item pairs. The complexity of item-item graph fusion is $O(|\mathcal{M}|^2|\mathcal{I}|^2)$ because structure-aware fusion is performed among multiple item-item graphs. **Each graph is frozen and remains unchanged during training**. Therefore, the above graph construction and fusion costs are incurred before training and are not repeatedly performed in each training epoch.

For the training stage, we report the following complexity at the epoch level by first analyzing the mini-batch cost and then multiplying it by the approximate number of batches, i.e., $|\mathcal{I}|/|B|$. We employ LightGCN as the convolutional kernel. Its complexity is $O(2\ell_1 d|E|\frac{|\mathcal{I}|}{|B|})$, where $|E|$ is the number of edges in the graph, ℓ_1 is the number of user-item graph convolutional layers, and $|B|$ is the batch size. The complexity for multimodal graph convolution is $O((|\mathcal{M}|+1)\ell_2 dk\frac{|\mathcal{I}|}{|B|})$, where ℓ_2 is the number of item-item graph convolutional layers.

In the HSC module, the complexity of the basic contrastive learning is $O(|\mathcal{M}||B||\mathcal{I}|d)$. The complexity of the differentiated contrastive learning is $O(|\mathcal{M}|\ |\mathcal{I}|\ d\ (|B|+k\ \log|B|))$.

In the LAD module, the computational cost mainly arises from the reverse denoising timesteps T and Transformer encoder with a complexity of $O(Td_h^2|\mathcal{I}|)$, where d_h represents the hidden dimension of the Transformer encoder. Through a synergistic design with contrastive alignment, the diffusion process can be effectively performed with a small number of timesteps T, thereby significantly reducing the theoretical time complexity.

4. Experiment

In this section, we conduct comprehensive experiments to answer the following research questions:

RQ1: How effective is our SIGNAL compared with the state-of-the-art general recommendation methods and MRs?

RQ2: How does each component of SIGNAL contribute to the overall performance?

RQ3: How sensitive is SIGNAL with different hyperparameters?

RQ4: How does the synergistic design of SIGNAL promote multimodal sematics learning?

RQ5: How does SIGNAL quantitatively outperform SOTA methods under varying sparsity and noise conditions?

4.1. Experimental settings

4.1.1. Datasets

To evaluate our proposed model, we conduct experiments on three widely adopted Amazon datasets following previous works (Li et al., 2026a; Liu et al., 2025; Zhang et al., 2021; Zhou et al., 2023c): Baby, Sports, and Clothing. Consistent with Guo et al. (2024), Xv et al. (2024), Zhou and Shen (2023), Zhou et al. (2023c), we apply 5-core filtering to both users and items, and utilize publicly released 4096-dimensional visual features and 384-dimensional textual features.

To further test the effectiveness of SIGNAL with various modalities and the usability in various situations, we evaluate it on the TikTok dataset, as used in Cheng and Song (2025), Jiang et al. (2024), Li et al. (2025a), Wei et al. (2023). This dataset is collected from the TikTok platform, which includes visual, textual, and acoustic modalities of items and allows us to examine the effectiveness of our approach across diverse input types. Table 3 provides a detailed summary of these datasets.

4.1.2. Evaluation metrics

For a fair performance evaluation, we randomly partition the historical interactions for each dataset into training, validation, and test sets using an 8:1:1 ratio, following Zhou et al. (2023c). The performance of the top-K recommendations is assessed using two widely adopted metrics: Recall@K (R@K) and Normalized Discounted Cumulative Gain@K (NDCG@K). Results are reported as user-averaged values on the test set for $K = 10$ and $K = 20$. Higher metric values indicate superior model performance.

4.1.3. Baselines

We evaluate SIGNAL against three categories of baseline: (1) General models without using multimodal features, (2) Graph-based and CL-enhanced MRs, (3) Diffusion-based MRs.

Table 3
Statistics of four real-world datasets with Visual(V), Textual(T), and Acoustic(A) contents of items.

Dataset	Baby		Sports		Clothing		TikTok		
Modality	V	T	V	T	V	T	V	T	A
Embed Dim	4096	384	4096	384	4096	384	128	768	128
User	19,445		35,598		39,387		9,308		
Item	7,050		18,357		23,033		4,710		
Interactions	160,792		296,337		278,677		68,722		
Sparsity	99.88%		99.95%		99.97%		99.89%		

(1) General models without using multimodal features
MF-BPR (Rendle et al., 2009) leverages the BPR loss to learn user preferences based on matrix factorization.
LightGCN (He et al., 2020) simplifies graph convolution by removing feature transformations and non-linear activations for efficient embeddings learning.
(2) Graph-based and CL-enhanced MRs
MMGCN (Wei et al., 2019) constructs separate user-item graph for each modality and uses graph convolution to capture modality-specific user preferences.
LATTICE (Zhang et al., 2021) explores semantic relationships between items by building modality-aware item–item graphs based on item similarities.
MICRO (Zhang et al., 2022) leverages graph structure to uncover latent item relationships from multimodal features and devises a contrastive learning framework to integrate multimodal relational signals.
BM3 (Zhou et al., 2023c) enhances representation learning by bootstrapping latent representations through a dropout-based contrastive view generation and jointly optimizing multi-modal objectives without negative samples.
MGCN (Yu et al., 2023) captures multimodal correlations via item-item graphs and employs a behavior-guided contrastive learning task to align collaborative and multimodal views.
FREEDOM (Zhou & Shen, 2023) reduces computational cost and boost recommendation accuracy by freezing the item–item graph while simultaneously denoising the user–item graph.
DA-MRS (Xv et al., 2024) handles noisy multi-modal content and user feedback by denoising item-item graphs and aligning multi-modal features with user preferences.
LGMRec (Guo et al., 2024) captures both local and global user interests by local GNNs and hypergraph learning before fusing them contrastive framework for robust recommendations.
NEGCL (Shi et al., 2025) boosts performance by incorporating random noise into multimodal contrastive learning to promote embeddings uniformity and utilizing an attention mechanism to weight user preferences adaptively.
(3) Diffusion-based MRs
DiffMM (Jiang et al., 2024) introduces the diffusion model to reconstruct user-item interaction graphs across different modalities and combines cross-modal contrastive learning to align multi-modal features with user-item interactions.
MoDiCF (Li et al., 2025a) generates fair and accurate recommendations by diffusing missing modalities from learned distributions and mitigating visibility bias through counterfactual inference in incomplete multimodal scenarios.

4.1.4. Implementation details

The proposed SIGNAL is implemented by PyTorch framework on a NVIDIA RTX A40 GPU with 48 GB memory. Simultaneously, to ensure a fair comparison, we employ a default batch size of 4096, learning rate of 0.001, and regularization coefficient λ_r of 0.01, with all baselines performing grid search according to the original papers. All main experiments and analyses are conducted with a fixed random seed for consistent comparisons with the baselines. To assess stability, we additionally evaluate SIGNAL and SIGNAL-LAD over four random seeds, which affect embedding and SCE initialization, data shuffling, and LAD noise sampling.

For graph construction, we strictly avoid using validation and test interactions. The co-occurrence graph is constructed only from the user-item interactions in the training set. For visual, textual, and acoustic modality graphs, we compute item-item similarities using only the corresponding item side features, which are available for the candidate item set and do not contain user-item interaction labels. Therefore, these content-based modality graphs do not use validation or test interactions, while all interaction-dependent graph structures are built from training data only.

In SIGNAL, the user-item graphs are processed by 2-layer convolutions in LightGCN ($\ell_2 = 2$), and the item-item graphs are processed by 1-layer convolution ($\ell_1 = 1$). The pruning threshold ξ is set to 2, and the early stopping strategy is applied with a patience of 25 epochs using the Adam optimizer (Kingma & Ba, 2014). Against hyperparameters, the number k of item neighbors is selected from the set $\{5, 10, 15, 20\}$, the semantic pre-alignment weight α_1 for multimodal features is selected from the set $\{0.3, 0.4 \ldots, 0.7\}$, and the ID semantic pre-alignment weight α_2 is selected from the set $\{0.05, 0.1, 0.15, 0.2\}$. Besides, the contrastive loss weights λ_{cl}^b and λ_{cl}^d are both selected from the set $\{1, 1e^{-1}, \ldots, 1e^{-4}\}$, the multimodal similarity threshold ρ is selected from the set $\{4, 5, \ldots, 8\}$, the diffusion timesteps T is selected from the set $\{0, 1, 2, 3, 4\}$, the residual weight μ is selected from the set $\{0.4, 0.5, \ldots, 0.8\}$, and the diffusion loss weight λ_{dm} is selected from the set $\{10, 1, \ldots, 1e^{-3}\}$. The optimal hyperparameter configuration is selected through grid search on the validation set. In the analysis of the synergy between CL and LAD, we additionally evaluate larger diffusion timesteps, i.e.,$T = 10$, $T = 20$ and $T = 50$, only to study the effect of multi-step denoising. These larger timestep settings are not used for selecting the main experimental results. Further implementation details and source code are available at https://github.com/Sail-high/SIGNAL.

4.2. Performance comparison (RQ1)

The overall fixed-seed performance of all baselines and our SIGNAL on four datasets is shown in Table 4. To further examine the robustness of the proposed model, Table 5 additionally reports the mean and standard deviation of SIGNAL and SIGNAL-LAD over four random seeds. From the two tables, we can observe that:

(1) Initially, compared with general models that do not utilize multimodal features, SIGNAL and most MRs demonstrate significant performance improvements. This highlights the effectiveness of fusing multimodal features, as they mitigate interaction sparsity by enriching item representations. Crucially, however, the performance of MMGCN (Wei et al., 2019) is partially weaker than that of LightGCN (He et al., 2020). This is attributed to the fact that the direct fusion of multimodal features often introduces excessive noise, which highlights the critical need for effective noise mitigation in multimodal recommendation.
(2) Building upon this necessity, SIGNAL achieves the best performance under the fixed-seed comparison protocol used in Table 4. Specifically, SIGNAL improves the best baseline by 5.54%, 8.37%, 5.10%, and 6.10% in terms of Recall@20 on the Baby, Sports, Clothing, and TikTok datasets, respectively. Table 5 further shows that SIGNAL maintains stable performance over four random seeds, with small standard deviations on most datasets and metrics. This indicates that the improvements of the proposed model are not caused by a specific random initialization. These gains are attributed to holistic semantic learning through PGE-HSC synergy and nuanced cross-modal conflict refinement through LAD.

Table 4
Overall performance comparison of SIGNAL and other baselines on four datasets. The best result is in boldface and the second best is underlined (without considering SIGNAL-LAD). "-" indicates that the model runs out of memory on an NVIDIA RTX A40 GPU with 48 GB memory, and therefore the corresponding result cannot be reported.

Dataset	Baby				Sports				Clothing				TikTok			
Model	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**
MF-BPR	0.0357	0.0575	0.0192	0.0249	0.0432	0.0653	0.0241	0.0298	0.0187	0.0279	0.0103	0.0126	0.0177	0.0346	0.0073	0.0130
LightGCN	0.0479	0.0754	0.0257	0.0328	0.0569	0.0864	0.0311	0.0387	0.0340	0.0526	0.0188	0.0236	0.0413	0.0653	0.0224	0.0282
MMGCN	0.0378	0.0615	0.0200	0.0261	0.0370	0.0605	0.0193	0.0254	0.0218	0.0345	0.0110	0.0142	0.0525	0.0730	0.0251	0.0307
LATTICE	0.0547	0.0850	0.0292	0.0370	0.0620	0.0953	0.0335	0.0421	0.0492	0.0733	0.0268	0.0330	0.0558	0.0843	0.0309	0.0367
MICRO	0.0577	0.0919	0.0314	0.0405	0.0676	0.1039	0.0365	0.0463	0.0506	0.0752	0.0274	0.0337	0.0561	0.0939	0.0314	0.0413
BM3	0.0564	0.0883	0.0301	0.0383	0.0656	0.0980	0.0355	0.0438	0.0422	0.0621	0.0231	0.0281	0.0643	0.0939	0.0325	0.0399
MGCN	0.0623	0.0956	0.0332	0.0418	0.0729	0.1081	0.0403	0.0488	0.0626	0.0929	0.0332	0.0410	0.0578	0.0953	0.0329	0.0437
FREEDOM	0.0627	0.0992	0.0330	0.0424	0.0717	0.1089	0.0385	0.0481	0.0629	0.0941	0.0341	0.0420	0.0561	0.0942	0.0312	0.0408
LGMRec	0.0644	0.0985	0.0344	0.0433	0.0698	0.1053	0.0380	0.0472	0.0542	0.0815	0.0292	0.0361	0.0512	0.0860	0.0266	0.0353
DA-MRS	0.0612	0.0966	0.0326	0.0416	0.0712	0.1075	0.0390	0.0483	0.0619	0.0907	0.0334	0.0407	0.0686	0.1054	0.0378	0.0470
NEGCL	0.0648	0.0988	0.0350	0.0438	0.0733	0.1089	0.0405	0.0497	0.0601	0.0890	0.0332	0.0406	0.0525	0.0906	0.0259	0.0355
DiffMM	0.0626	0.0975	0.0322	0.0409	0.0671	0.1014	0.0363	0.0454	0.0522	0.0791	0.0288	0.0354	0.0674	0.1098	0.0345	0.0445
MoDiCF	0.0549	0.0879	0.0293	0.0371	–	–	–	–	–	–	–	–	0.0603	0.0943	0.0323	0.0407
SIGNAL-LAD	0.0667	0.1030	0.0353	0.0446	0.0768	0.1158	0.0415	0.0514	0.0669	0.0974	0.0362	0.0438	0.0683	0.1057	0.0356	0.0450
SIGNAL	**0.0679**	**0.1047**	**0.0368**	**0.0463**	**0.0783**	**0.1180**	**0.0424**	**0.0526**	**0.0674**	**0.0989**	**0.0367**	**0.0446**	**0.0772**	**0.1165**	**0.0408**	**0.0506**
Improv.	4.78%↑	5.54%↑	5.14%↑	5.71%↑	6.82%↑	8.37%↑	4.69%↑	5.84%↑	7.15%↑	5.10%↑	7.62%↑	6.19%↑	12.53%↑	6.10%↑	7.93%↑	7.65%↑

Table 5
Robustness analysis of SIGNAL and SIGNAL-LAD over four random seeds. The first line in each cell denotes the mean value, and the second line denotes the standard deviation.

Method	Baby				Sports				Clothing				TikTok			
	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20
SIGNAL-LAD	0.0663 ±0.0007	0.1025 ±0.0010	0.0354 ±0.0003	0.0445 ±0.0004	0.0765 ±0.0003	0.1155 ±0.0006	0.0415 ±0.0002	0.0512 ±0.0003	0.0668 ±0.0003	0.0973 ±0.0005	0.0362 ±0.0003	0.0439 ±0.0004	0.0702 ±0.0019	0.1070 ±0.0016	0.0367 ±0.0013	0.0465 ±0.0015
SIGNAL	**0.0681** ±0.0009	**0.1046** ±0.0012	**0.0368** ±0.0003	**0.0461** ±0.0003	**0.0786** ±0.0004	**0.1179** ±0.0008	**0.0422** ±0.0003	**0.0525** ±0.0004	**0.0672** ±0.0004	**0.0991** ±0.0006	**0.0365** ±0.0003	**0.0446** ±0.0004	**0.0770** ±0.0014	**0.1162** ±0.0018	**0.0405** ±0.0011	**0.0503** ±0.0013

(3) Furthermore, compared with other graph-based and contrastive-learning-enhanced MRs, the SIGNAL-LAD variant, which retains only the graph encoding and contrastive components, still achieves substantial performance improvements. This demonstrates that PGE and HSC provide a strong foundation for holistic semantic learning. This superior enhancement stems from our key designs: During semantic encoding, unlike methods such as LATTICE (Zhang et al., 2021) and FREEDOM (Zhou & Shen, 2023), PGE additionally incorporates a behavior-aware co-occurrence graph, denoises modality-specific graphs, and promotes cross-modal information interaction through inter-graph fusion, thereby narrowing representation gaps during graph propagation. During semantic alignment, unlike MICRO (Zhang et al., 2022) and MGCN (Yu et al., 2023), HSC differentiates item relations according to graph-encoded neighborhoods and cross-view semantic consistency, thereby preserving the encoded semantic structure. Table 5 further shows that SIGNAL-LAD remains stable across different random seeds, confirming the robustness of the PGE–HSC combination.

(4) Finally, SIGNAL demonstrates consistent effectiveness across all four datasets. On Baby, Sports, and Clothing, SIGNAL-LAD performs close to the full model, suggesting that PGE and HSC already resolve a substantial portion of the semantic discrepancies through holistic encoding and alignment. Nevertheless, the additional improvements of full SIGNAL confirm that LAD provides complementary partial refinement rather than merely duplicating the effects of holistic alignment. The contribution of LAD is particularly pronounced on TikTok, which is consistent with the heterogeneous visual, textual, and acoustic signals that leave greater scope for refining residual cross-modal conflicts. Moreover, although LAD introduces stochasticity through diffusion-based restoration, Table 5 shows that its average performance gains exceed the corresponding fluctuations across random seeds. We further examine the necessity of LAD in Section 4.3.2 and its cost–benefit trade-off in Section 4.6.1.

4.3. Ablation study (RQ2)

4.3.1. Core component ablation

To evaluate the effectiveness of each component in SIGNAL, we designed five variants for ablation studies: (1) **SIGNAL-ACL**, which removes all contrastive learning mechanisms; (2) **SIGNAL-BCL**, which removes the basic contrastive learning component, meaning it no longer performs the holistic semantic alignment across modalities; (3) **SIGNAL-DCL**, which removes the differentiated contrastive learning module, thereby failing to preserve the encoded semantic structure during alignment; (4) **SIGNAL-LAD**, which removes the lightweight attentional diffusion module, thus failing to handle nuanced conflicts of different modalities and (5) **SIGNAL-PGE**, which removes the designed item-item graph fusion step from the pre-aligned graph encoding module, and retains the other graph construction, denoising and encoding operations for modality features, resulting in the absence of pre-alignment between multimodal and ID features of items. Results in Fig. 6 reveal the following key findings.

(1) The effectiveness of the proposed components is confirmed, as the ablation of each component results in a significant performance drop across all benchmarks. This degradation is particularly pronounced on the TikTok dataset, underscoring that in highly complex multimodal scenarios, the combined efforts of all components are important for maintaining high performance.

(2) Contrastive learning is essential for effective multimodal semantic learning. Removing **BCL** causes a clear performance decline, which is most pronounced on TikTok. This is consistent with its richer

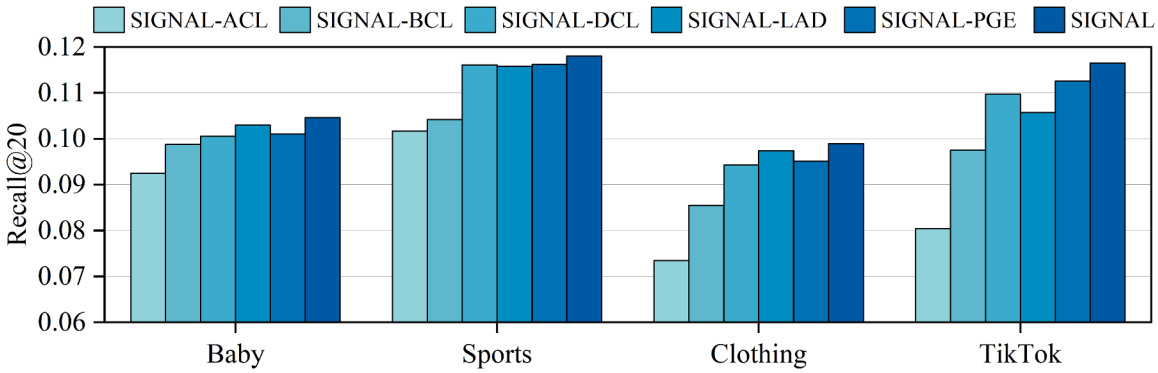


Fig. 6. Ablation of different components on SIGNAL.

visual, textual, and acoustic signals, for which BCL plays an important role in bridging holistic semantic gaps across modalities. Moreover, without BCL, LAD loses the holistically aligned initialization required for efficient partial conflict refinement, weakening the intended synergy between the two modules. Removing **DCL** also leads to noticeable degradation, with Recall@20 decreasing by 4.0%, 1.6%, 4.7%, and 5.8% on the four datasets, respectively. Compared with BCL, DCL provides a more structure-oriented benefit by preserving the semantic neighborhoods established by PGE during contrastive alignment. Maintaining these meaningful item relations prevents semantically related items from being indiscriminately separated and provides a more reliable semantic structure for accurate recommendation.

(3) The **LAD** module plays an important role in partial-granularity semantic alignment by mitigating nuanced cross-modal conflicts. Removing LAD leads to a clear performance decline in the **SIGNAL-LAD** variant. This is because some local cross-modal inconsistencies remain after graph-based pre-alignment and contrastive semantic alignment and therefore require further refinement. While holistic alignment cannot explicitly resolve these nuanced conflicts, LAD models them as semantic interference and restores conflicting features under reliability-aware guidance. This effect is most evident on TikTok, where Recall@20 decreases from 0.1165 to 0.1057 after removing LAD. The larger decline is consistent with the richer visual, textual, and acoustic signals in TikTok, which leave greater scope for refining residual cross-modal conflicts.

(4) The **PGE** module contributes to holistic semantic learning by establishing cross-modal structural interactions. In the **SIGNAL-PGE** variant, we remove inter-graph fusion while retaining modality-specific graph construction and encoding. Therefore, the resulting performance decline directly reflects the contribution of adaptive graph fusion. Specifically, Recall@20 decreases by 3.5%, 3.8%, and 3.3% on Baby, Clothing, and TikTok, respectively. These results indicate that the modality-specific graphs in these datasets contain complementary and inconsistent structural information, for which inter-graph attention adaptively weights cross-modal neighborhood consistency to promote more effective semantic interaction. The smaller decline of 1.5% on Sports suggests greater structural consistency across its modality graphs, leaving less room for adaptive fusion to provide additional benefits.

4.3.2. Necessity of LAD

To evaluate the necessity of LAD for partial conflict refinement, we compare it with several alternative designs. **MLP-Calib** constructs discrepancy-aware features from the item ID and modality-specific representations, and uses an MLP to predict a residual vector for one-step calibration of the averaged multimodal representation. **Attn-Refine** uses the item ID representation as the query and modality-specific representations as keys and values to perform reliability-aware modality aggregation. The $\mathbf{LAD}_{\mathrm{GN}}$ variant retains the degradation–restoration architecture of LAD but replaces modality-mixed semantic noise with standard Gaussian noise, while **w/o Refine** removes partial conflict refinement entirely. All variants share the same PGE and HSC components and differ only in the refinement strategy or noise design.

Table 6
Effectiveness analysis of diffusion-based partial conflict refinement and modality-mixed semantic noise. The best results are highlighted in bold and the second-best results are underlined.

Variant	Baby		Sports		Clothing		TikTok	
	R@20	N@20	R@20	N@20	R@20	N@20	R@20	N@20
w/o Refine	0.1030	0.0446	0.1158	0.0514	0.0974	0.0438	0.1057	0.0450
MLP-Calib	0.1022	0.0445	<u>0.1165</u>	0.0518	0.0971	0.0435	0.1074	0.0459
Attn-Refine	<u>0.1039</u>	<u>0.0457</u>	0.1163	<u>0.0522</u>	<u>0.0980</u>	<u>0.0442</u>	<u>0.1108</u>	<u>0.0482</u>
$\mathrm{LAD}_{\mathrm{GN}}$	0.1034	0.0449	0.1161	0.0516	0.0978	0.0440	0.1081	0.0459
$\mathrm{LAD}_{\mathrm{Ours}}$	**0.1047**	**0.0463**	**0.1180**	**0.0526**	**0.0989**	**0.0446**	**0.1165**	**0.0506**

As shown in Table 6, LAD achieves the best performance across all datasets and metrics. Compared with w/o Refine, MLP-Calib provides limited gains and occasionally causes performance degradation, indicating that one-step residual calibration is insufficient for reliably handling complex cross-modal discrepancies. Attn-Refine generally performs better than MLP-Calib, demonstrating the benefit of reliability-aware modality weighting. However, it remains inferior to LAD, suggesting that reweighting existing modality representations alone cannot fully resolve nuanced semantic conflicts. By restoring conflict-related features under reliability-aware guidance, LAD provides more effective partial refinement than both one-step calibration and attention-only aggregation.

We further compare $\mathbf{LAD}_{\mathrm{GN}}$ and $\mathbf{LAD}_{\mathrm{Ours}}$ to evaluate the effectiveness of the proposed modality-mixed semantic noise. Although standard Gaussian-noise diffusion provides modest improvements, $\mathbf{LAD}_{\mathrm{Ours}}$ consistently performs better across all datasets and metrics. On TikTok, modality-mixed semantic noise improves R@20 and N@20 by 7.78% and 10.24%, respectively, over standard Gaussian noise. These results support that item-dependent perturbations constructed from modality-specific representations better reflect cross-modal interference than content-independent element-wise Gaussian noise. The benefit of LAD is particularly pronounced on TikTok. This pattern is consistent with its heterogeneous visual, textual, and acoustic signals, which leave greater scope for refining residual cross-modal discrepancies after holistic alignment.

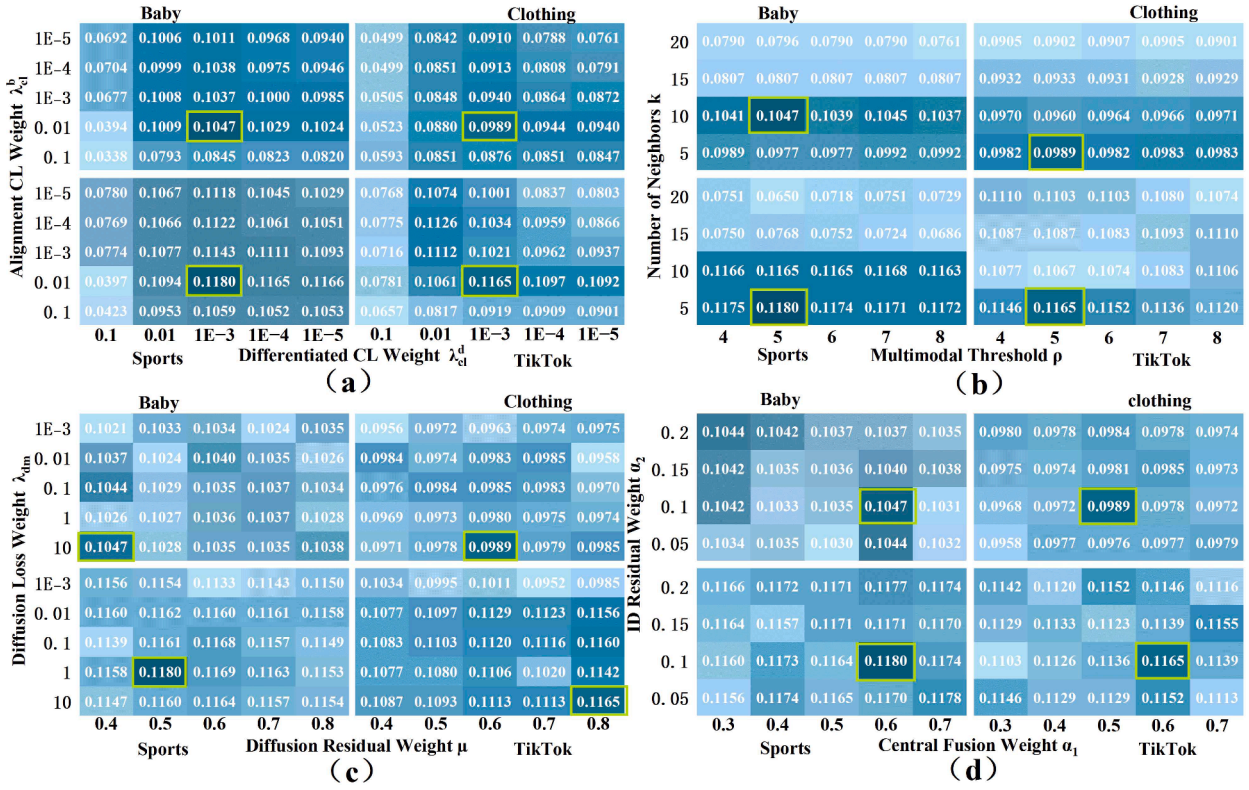


Fig. 7. Sensibility analysis: the intensity of color change represents the sensitivity.

4.4. Hyperparametric sensibility analysis(RQ3)

We explore the impacts of all hyperparameters on performance across datasets. The results are presented in Fig. 7, where the intensity of color change represents the sensitivity and the optimal result is highlighted by a green circle. From this figure, we can observe that:

The Pair of Hyperparameter (λ_{cl}^b, λ_{cl}^d). The hyperparameter λ_{cl}^b governs the semantic alignment, while λ_{cl}^d regulates the semantic discrimination. During the contrastive process, these two parameters compete in shaping the semantic structure of samples. Improper configuration of this parameter pair can significantly distort the learned semantic relationships; hence, SIGNAL exhibits high sensitivity to their settings in Fig. 7 (a).

The Pair of Hyperparameter (k, ρ). The hyperparameter k controls the number of neighbors for each item in the graph structure and also influences structural fusion, while ρ represents the multimodal similarity threshold. Experimental results on all four datasets indicate that SIGNAL achieves stronger performance with relatively small k values, as larger k tends to cause over-smoothing of the convolved features. Meanwhile, SIGNAL shows insensitivity to the value of ρ in Fig. 7 (b), as the relatively high threshold imposes strict selection criteria, ensuring that minor fluctuations in ρ do not significantly alter the sample classification outcomes.

The Pair of Hyperparameter (λ_{dm}, μ). The hyperparameter λ_{dm} controls the weight of the LAD training loss, while μ regulates the residual connections between the pre-diffusion and post-diffusion embeddings. SIGNAL exhibits low sensitivity to this hyperparameter pair in Fig. 7(c); however, we observe during training that a larger λ_{dm} leads to faster convergence.

The Pair of Hyperparameter (α_1, α_2). The hyperparameter α_1 controls the weight of multimodal semantic pre-alignment, while α_2 governs the weight of collaborative ID embeddings pre-alignment. Excessively large values cause features to become over-aligned, losing their distinctive characteristics, whereas overly small values lead to insufficient alignment. An intermediate range of values yields the best performance in Fig. 7 (d).

Table 7
Optimal parametric setting for four datasets.

Parameters	λ_{cl}^b	λ_{cl}^d	k	ρ	μ	λ_{dm}	α_1	α_2
Baby	0.01	0.001	10	5	0.4	10	0.6	0.1
Sports	0.01	0.001	5	6	0.5	1	0.5	0.1
Clothing	0.01	0.001	5	5	0.6	10	0.5	0.1
TikTok	0.01	0.001	5	5	0.8	10	0.6	0.1

Based on the results and analysis in Fig. 7, we summarize the optimal hyperparameters sampled for each dataset in Table 7.

4.5. Synergistic analysis (RQ4)

4.5.1. GNNs and CL synergy analysis

To comprehensively validate the synergistic design between GNNs and CL, we present an empirical analysis from three perspectives: performance universality, semantic evolution, and optimization convergence.

Performance Synergy and Cross-Backbone Compatibility. We integrate PGE and DCL individually and jointly into two representative GNNs- and CL-based methods, MICRO (Zhang et al., 2022) and MGCN (Yu et al., 2023), as well as our own backbone. For our backbone, LAD is removed to isolate the interaction between PGE and DCL and minimize the influence of other components. Table 8 provides two main observations.

(1) Individually incorporating PGE or DCL generally improves all three backbones across the four datasets, demonstrating their compatibility with different graph encoding and contrastive learning frameworks. Their joint integration achieves the strongest overall performance, including the best Recall@20 on every dataset, with relative improvements ranging from 2.31% to 9.99% over the original backbones. On our backbone, the joint variant improves Recall@20 by

Table 8
Compatibility and synergistic analysis of our proposed modules (PGE and DCL) integrated into representative GNNs + CL methods across four datasets. "Improv." indicates the relative improvement over the original base model.

Dataset	Baby				Sports				Clothing				TikTok			
Model	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**
MICRO	0.0577	0.0919	0.0314	0.0405	0.0676	0.1039	0.0365	0.0463	0.0506	0.0752	0.0274	0.0337	0.0561	0.0939	0.0314	0.0413
w/ PGE	0.0588	0.0940	0.0321	0.0413	0.0692	0.1045	0.0374	0.0470	0.0519	0.0766	0.0280	0.0344	0.0581	0.0967	0.0323	0.0435
w/ DCL	0.0594	0.0941	0.0318	0.0412	0.0689	0.1051	0.0374	0.0468	0.0515	0.0765	0.0279	0.0342	0.0583	0.0951	0.0329	0.0429
w/ PGE & DCL	0.0602	0.0953	0.0324	0.0417	0.0699	**0.1063**	0.0377	0.0475	0.0523	0.0772	0.0283	0.0348	0.0595	**0.0997**	0.0335	0.0447
Improv.	4.33%↑	3.70%↑	3.18%↑	2.96%↑	3.40%↑	2.31%↑	3.29%↑	2.59%↑	3.36%↑	2.66%↑	3.28%↑	3.26%↑	6.06%↑	6.18%↑	6.69%↑	8.23%↑
MGCN	0.0623	0.0956	0.0332	0.0418	0.0729	0.1081	0.0403	0.0488	0.0626	0.0929	0.0332	0.0410	0.0578	0.0953	0.0329	0.0437
w/ PGE	0.0640	0.0971	0.0337	0.0424	0.0738	0.1102	0.0408	0.0496	0.0640	0.0939	0.0340	0.0425	0.0585	0.0993	0.0342	0.0458
w/ DCL	0.0633	0.0970	0.0339	0.0425	0.0744	0.1096	0.0409	0.0499	0.0646	0.0949	0.0340	0.0426	0.0582	0.0988	0.0341	0.0455
w/ PGE & DCL	0.0645	**0.0989**	0.0342	0.0429	0.0751	**0.1124**	**0.0415**	**0.0509**	0.0653	0.0958	0.0347	0.0429	**0.0599**	0.1023	0.0351	0.0468
Improv.	3.53%↑	3.45%↑	3.01%↑	2.63%↑	3.02%↑	3.98%↑	2.98%↑	4.30%↑	4.31%↑	3.12%↑	4.52%↑	4.63%↑	3.63%↑	7.34%↑	6.69%↑	7.09%↑
Ours	0.0627	0.0958	0.0324	0.0411	0.0733	0.1102	0.0398	0.0491	0.0633	0.0896	0.0346	0.0415	0.0622	0.0961	0.0327	0.0433
w/ PGE	0.0651	0.0996	0.0346	0.0436	0.0752	0.1128	0.0409	0.0511	0.0659	0.0940	0.0352	0.0430	0.0655	0.0990	0.0339	0.0452
w/ DCL	0.0652	0.0991	0.0332	0.0429	0.0757	0.1130	0.0417	0.0503	0.0654	0.0938	0.0354	0.0429	0.0661	0.1016	0.0344	0.0450
w/ PGE & DCL	0.0667	0.1030	0.0353	0.0446	0.0768	0.1158	0.0415	0.0514	0.0669	0.0974	0.0362	0.0438	0.0683	0.1057	0.0356	0.0450
Improv.	6.38%↑	7.52%↑	8.95%↑	8.52%↑	4.77%↑	5.08%↑	4.27%↑	4.68%↑	5.69%↑	8.71%↑	4.62%↑	5.54%↑	9.81%↑	9.99%↑	8.87%↑	3.93%↑

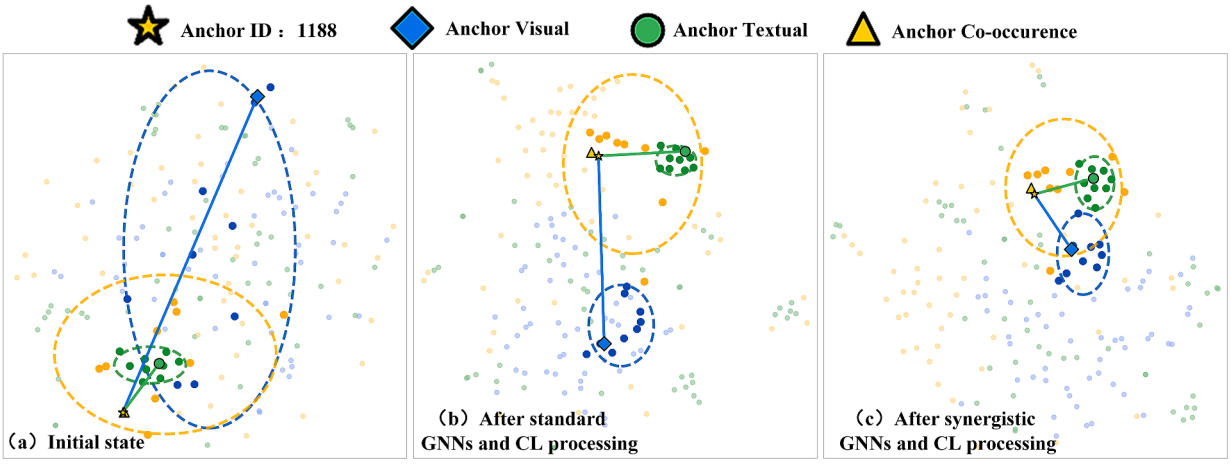


Fig. 8. Visualization of embedding distributions on Baby. Black-outlined points denote the anchor item's ID and multimodal embeddings, where different colors indicate different views. The semantic neighborhoods formed by the Top-10 similar items are highlighted with dashed circles. Low-opacity points represent randomly sampled background items.

7.52%, 5.08%, 8.71%, and 9.99% on Baby, Sports, Clothing, and TikTok, respectively. Although the additional gain over the stronger single-module variant varies across datasets and backbones, it remains positive in all cases. These results support the cross-backbone applicability of PGE and DCL and show that their joint use provides consistent complementary benefits.

(2) These joint benefits arise from the complementary roles of PGE and DCL across the encoding and alignment stages. PGE reduces isolated modality-specific encoding by initiating preliminary cross-modal alignment during graph propagation, thereby providing a more favorable starting point for subsequent contrastive alignment. In turn, DCL distinguishes item relations according to the semantic neighborhoods established during graph encoding, reducing the indiscriminate repulsion of semantically related items and preserving the encoded neighborhood structure. Together, PGE improves the representations entering contrastive alignment, while DCL protects the semantic relations established by PGE, thereby supporting the proposed bidirectional encoding–alignment synergy.

Visualization of Semantic Evolution. We further employ t-SNE (van der & Hinton, 2008) on the Baby dataset to qualitatively illustrate how the proposed synergy affects the embedding distributions. Specifically, we randomly select an anchor item (ID: 1188) and visualize its ID embedding together with its multimodal embeddings, where different colors denote different views and black outlines highlight the anchor points. For each view, we further identify the Top-10 similar items as neighborhood references and randomly sample 50 items from ranks 10–510 as low-opacity background points. The Top-10 neighborhoods are highlighted by dashed circles for visual comparison. As shown in Fig. 8, three observations can be made.

(1) In the initial state, the anchor representations exhibit large and irregular cross-view gaps. The visual and co-occurrence views fail to form compact and coherent neighborhoods around their corresponding anchors.

(2) After standard GNNs and CL processing, the multimodal anchor embeddings move closer to the anchor ID embedding, indicating improved holistic semantic alignment. Meanwhile, neighboring items in each view begin to form local clusters, showing that GNNs-based encoding captures meaningful semantic proximity.

(3) After the synergistic processing of PGE and HSC, the Top-10 neighboring items form more compact semantic neighborhoods, particularly in the visual and co-occurrence views. This pattern is consistent with the intended neighborhood-preserving effect of DCL during

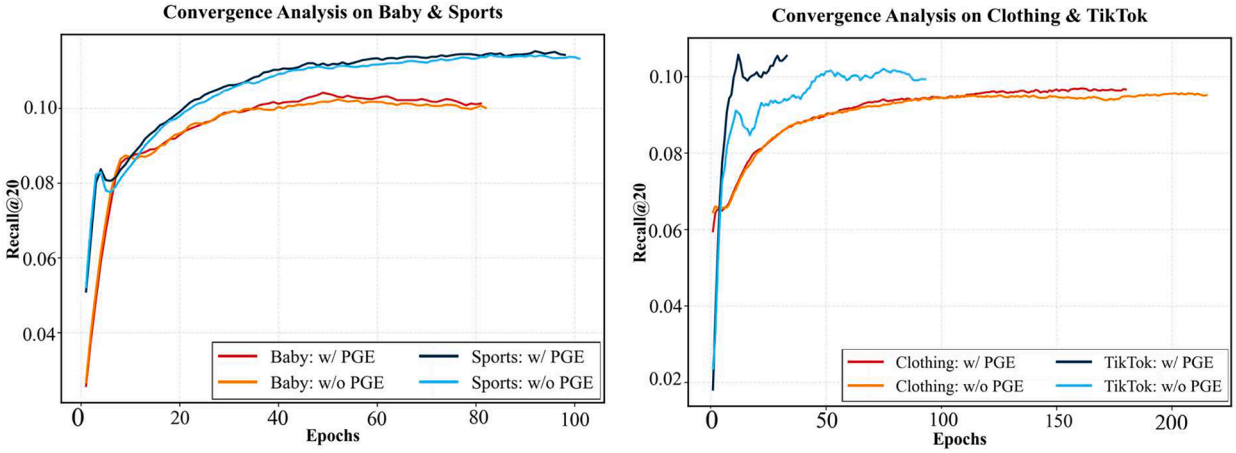


Fig. 9. Convergence analysis of PGE across four datasets. The curves illustrate the Recall@20 performance over epochs.

contrastive alignment. Meanwhile, the multimodal anchor embeddings become more consistent with the anchor ID embedding, more clearly reflecting the effect of PGE in narrowing cross-modal representation gaps through preliminary alignment during graph encoding. Overall, the visualization mainly supports the contribution of PGE to holistic alignment, while providing complementary qualitative evidence for the neighborhood-preserving role of DCL.

Visualization of Convergence. To verify the theoretical analysis that PGE provides a superior initialization for contrastive alignment, we conducted a convergence analysis by plotting the Recall@20 curves throughout the training process. To strictly isolate the contribution of the pre-alignment mechanism within the PGE module, we utilized the SIGNAL-LAD variant as the experimental baseline to minimize interference. We then compared the convergence behaviors of this variant **with and without pre-alignment**, denoted as **w/ PGE** and **w/o PGE**. As illustrated in Fig. 9, we observe two distinct advantages regarding optimization efficiency.

Firstly, the integration of pre-alignment contributes to improving the stability of the training process. For instance, around the 10th epoch, the variant w/ PGE on the Baby dataset appears more stable compared to variant w/o PGE. Similarly, on the Sports dataset, the introduction of pre-alignment results in reduced volatility. Secondly, pre-alignment facilitates a more rapid convergence process. On datasets such as Clothing and TikTok, the variants w/ PGE exhibit a steeper trajectory of performance growth, suggesting that they can achieve convergence at an earlier stage compared to those w/o PGE. These results indicate that the pre-aligned representations provide a favorable initialization for subsequent contrastive alignment, resulting in faster and more stable optimization.

4.5.2. CL and LAD synergy analysis

To analyze the synergy between holistic contrastive alignment and diffusion-based partial refinement, we examine whether BCL can reduce the number of diffusion steps required by LAD and whether LAD can further improve the representations after BCL. As evidenced by Fig. 10, we can observe that:

(1) With holistic alignment (solid markers) via BCL mechanism, merely one diffusion step (0 - 1) yields significant gains: R@20 increases from 0.1030 to 0.1047 (+1.65%) and N@20 from 0.0446 to 0.0463 (+3.81%) in Baby; Similarly in Sports, R@20 rises 1.90% (0.1158 to 0.1180) and N@20 rises 2.33% (0.0514 to 0.0526). These results confirm that LAD facilitates effective partial-granularity alignment in just one step, with low time overhead, when built upon a foundation of holistic alignment established by BCL.

(2) Without holistic alignment (hollow markers), LAD fails to deliver meaningful improvement in early steps, requiring costly multi-step (10 - 50) iterations for suboptimal alignment. As the number of steps increases, the computational cost grows substantially. These findings validate the synergistic effect achieved by combining BCL and LAD: while BCL establishes a foundational holistic alignment among modalities, LAD mitigates partial nuanced conflicts among modalities that contrastive alignment alone cannot handle, and this holistic alignment in turn accelerates the diffusion process in LAD.

These results indicate that the synergy between BCL and LAD lies in both effectiveness and efficiency: BCL provides a holistically aligned semantics that reduces the denoising burden of LAD, while LAD complements BCL by refining partial-granularity conflicts that holistic contrastive alignment cannot fully address.

4.6. Efficiency analysis

4.6.1. Time and parameter efficiency

To evaluate the efficiency of SIGNAL, we analyze its total per-epoch runtime, calculated as the sum of training and evaluation time, the number of learnable parameters, and the cost–benefit trade-off introduced by LAD. Runtime is measured by averaging the per-epoch time recorded in the logs. The parameter count includes user and item ID embeddings, multimodal item embeddings, and all trainable network layers. We further quantify the additional computational and tuning costs of LAD against its performance gains. The results are summarized in Table 9.

Computation Time. Without LAD, SIGNAL-LAD achieves competitive recommendation performance with low training and evaluation costs. Enabling LAD incurs a moderate runtime increase while providing further performance improvements. Compared with diffusion-based MRs such as DiffMM (Jiang et al., 2024) and MoDiCF (Li et al., 2025a), SIGNAL achieves a more favorable balance between recommendation accuracy and computational efficiency.

Learnable Parameters. Among graph-based and CL-enhanced MRs, LATTICE (Zhang et al., 2021) and FREEDOM (Zhou & Shen, 2023) contain a large number of learnable parameters. This is primarily due to the use of non-frozen multimodal features, which are either used to dynamically update graph structures (Zhang et al., 2021) or to enhance performance through multimodal loss (Zhou & Shen, 2023). Such designs often lead to increased model complexity and may slow down convergence. In contrast, SIGNAL and other MRs (LGMRec Guo et al., 2024, DA-MRS Xv et al., 2024, NEGCL Shi et al., 2025) freeze multimodal features, drastically reducing the number of trainable parameters. Among diffusion-based MRs (DiffMM Jiang et al., 2024, MoDiCF Li et al., 2025a), SIGNAL

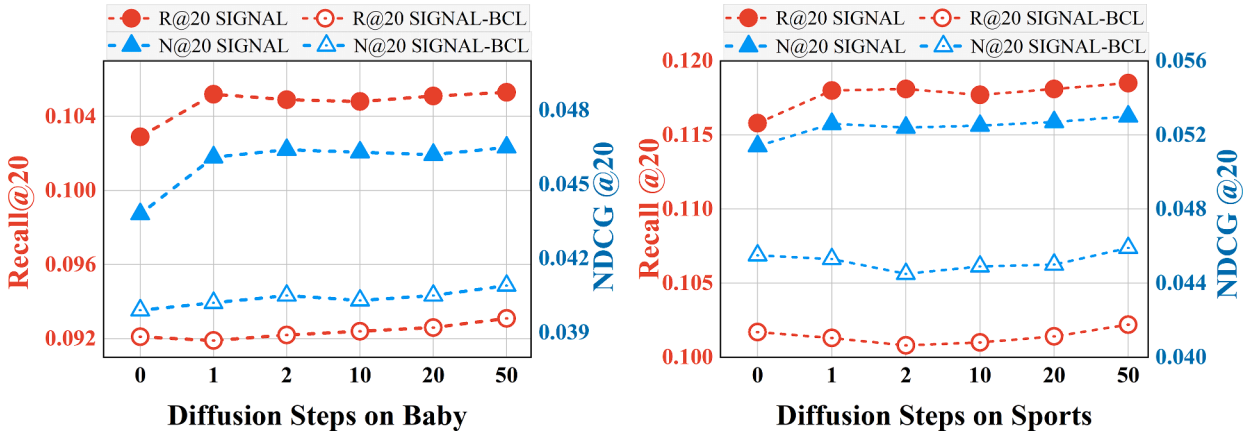


Fig. 10. Synergistic effect of BCL and LAD in SIGNAL on Baby and Sports.

Table 9
Comparison of our SIGNAL against baselines on efficiency. Parameter denotes the **learnable** parameter for the model, and Time denotes the total duration of each epoch, including both the **training** and **evaluation** phases. - indicates that the model runs out of memory on an NVIDIA RTX A40 GPU with 48 GB memory, so the corresponding result cannot be reported.

Dataset	Metrics	LATTICE	FREEDOM	LGMRec	DA-MRS	NEGCL	DiffMM	MoDiCF	SIGNAL-LAD	SIGNAL
Baby	Time(s/epoch)	3.93	3.20	3.80	4.89	6.55	9.91	89.2	3.91	4.57
	Parameter(M)	33.56	33.56	2.00	2.18	8.14	30.0	9.6	1.70	2.18
Sports	**Time(s/epoch)**	7.96	6.51	9.21	13.59	24.21	24.78	–	8.72	10.72
	Parameter(M)	85.98	85.98	3.76	3.74	15.17	77.2	–	3.45	3.93
Clothing	**Time(s/epoch)**	8.55	6.87	8.8	14.73	21.74	30.31	–	9.08	11.20
	Parameter(M)	107.47	107.47	4.30	4.28	17.34	96.5	–	3.99	4.47
TikTok	**Time(s/epoch)**	5.06	1.20	1.44	1.78	2.23	4.05	38.5	1.49	1.71
	Parameter(M)	7.96	7.96	1.09	1.08	4.53	41.1	7.8	1.03	1.51

demonstrates a clear parameter advantage. This is primarily attributable to the lightweight design of the LAD module. The core of LAD is implemented as a single-head, single-layer Transformer encoder, which consequently ensures that the module introduces only a moderate number of parameters to the overall SIGNAL architecture.

Cost–Benefit Trade-off of LAD. LAD introduces an additional SCE network and three hyperparameters, namely T, λ_{dm}, and μ. Compared with SIGNAL-LAD, it adds approximately 0.48M trainable parameters and accounts for 12.9%–18.9% of the total epoch time across the four datasets. In return, LAD improves Recall@20 by 1.65%, 1.90%, 1.54%, and 10.22% on Baby, Sports, Clothing, and TikTok, respectively.

The limited computational overhead benefits from the holistically aligned initialization provided by BCL, which allows LAD to refine residual conflicts with only a single restoration step ($T = 1$). The larger improvement on TikTok is consistent with its heterogeneous visual, textual, and acoustic signals, which leave greater scope for partial conflict refinement. Moreover, performance remains relatively stable over the evaluated ranges of λ_{dm} and μ, limiting the additional tuning burden. Together with the alternative-refinement comparisons in Section 4.3.2, these results show that LAD provides complementary conflict refinement with moderate computational and tuning costs.

4.6.2. Runtime and memory breakdown

Table 10 further separates the computational cost of SIGNAL into offline co-occurrence graph construction, multimodal graph preprocessing, and per-epoch training and evaluation. The co-occurrence graph is constructed once, saved as an item graph dictionary, and directly reused in subsequent experiments. The remaining preprocessing operations, including multimodal similarity computation, graph denoising, top-k graph construction, and inter-graph fusion, are performed before training.

Table 10
Runtime and memory breakdown of SIGNAL. Co-occurrence graph construction is a one-time offline preprocessing step, while the other preprocessing operations are performed before training. Training and evaluation time are reported per epoch.

Cost	Baby	Sports	Clothing	TikTok
Co-occurrence graph construction (s)	12	87	127	32
Multimodal similarity computation (s)	0.0457	0.2691	0.3688	0.0109
Graph denoising (s)	0.0063	0.0376	0.0596	0.0191
Top-k graph construction (s)	6.6139	19.8344	23.4501	6.7565
Inter-graph fusion (s)	0.1323	0.8727	1.3881	0.1989
Total on-the-fly preprocessing time (s)	6.8113	21.0301	25.3062	7.0096
Training time (s/epoch)	2.43	6.17	5.89	0.92
Evaluation time (s/epoch)	2.04	4.55	5.31	0.79
Peak GPU memory during training (MB)	5558	8100	11,601	3716
Sparse graph storage (MB)	11.3751	28.1622	33.1225	11.9520

The offline co-occurrence graph construction takes 12s, 87s, 127s, and 32s on Baby, Sports, Clothing, and TikTok, respectively. Although relatively expensive, this one-time cost does not affect per-epoch efficiency. The complete multimodal graph preprocessing requires 6.81s, 21.03s, 25.31s, and 7.01s on the four datasets. These totals include minor implementation overhead beyond the individually profiled components. Among them, top-k graph construction accounts for most of the runtime because neighbors must be selected for each item from the modality-specific similarity matrices, consistent with the theoretical complexity analysis.

SIGNAL requires 2.43s, 6.17s, 5.89s, and 0.92s per training epoch, and 2.04s, 4.55s, 5.31s, and 0.79s per evaluation epoch on the four datasets, respectively. Since graph construction is performed outside the epoch loop, its cost does not accumulate during training.

Table 11
Item-count scalability analysis on Clothing. Runtime growth is normalized by the 25% item subset to highlight the scaling trend, while memory consumption is reported in MB.

Item ratio	25%	50%	75%	100%
#Items	5758	11,516	17,275	23,033
Item count scale	1.00×	2.00×	3.00×	4.00×
Similarity computation time	1.00×	3.81×	8.90×	16.07×
Graph denoising time	1.00×	3.65×	8.23×	14.65×
Top-k graph construction time	1.00×	2.23×	4.45×	7.14×
Inter-graph fusion time	1.00×	3.84×	8.61×	15.20×
Total preprocessing time	1.00×	2.34×	4.74×	7.71×
Dense similarity storage (MB)	252.95	1011.80	2276.81	4047.54
Dense similarity storage scale	1.00×	4.00×	9.00×	16.00×
Peak preprocessing GPU memory (MB)	1293.94	3400.89	6837.96	11601.55
Sparse graph storage (MB)	8.03	16.25	24.64	33.12

The peak training GPU memory is 5558MB, 8100MB, 12320MB, and 3716MB, while the persistent sparse graph storage is only 11.38MB, 28.16MB, 33.12MB, and 11.95MB. Although multiple item graphs are constructed, they are stored in sparse top-k format, and the temporary dense similarity matrices are discarded after preprocessing.

Overall, the main computational cost of SIGNAL arises from one-time graph construction, particularly co-occurrence graph construction and top-k neighbor selection. Once the sparse graphs are obtained, the training, evaluation, and persistent storage costs remain manageable.

4.6.3. Item-count scalability

To analyze the scalability of SIGNAL with respect to item count, we conduct a controlled experiment on Clothing using 25%, 50%, 75%, and 100% of the items. Runtime growth is normalized by the 25% subset, while memory consumption is reported in MB. Co-occurrence graph construction is excluded because its cost also depends on the user–item interaction distribution; its full offline cost is reported separately in Table 10.

Runtime scalability. As shown in Table 11, increasing the item count from 1.00× to 4.00× results in 16.07×, 14.65×, and 15.20× increases in similarity computation, graph denoising, and inter-graph fusion time, respectively. These trends are consistent with the quadratic dependence of dense pairwise item operations on the number of items. However, the absolute runtime breakdown in Table 10 shows that top-k neighbor construction remains the dominant preprocessing component. Its runtime increases by 7.14×, while the total preprocessing time increases by 7.71×. Therefore, dense pairwise operations determine the asymptotic growth, whereas neighbor selection constitutes the main practical runtime overhead.

Memory scalability. Dense similarity storage increases from 252.95MB to 4047.54MB as the item count grows by 4.00×, exhibiting the expected 16.00× quadratic growth. Accordingly, peak preprocessing GPU memory increases from 1293.94MB to 11601.55MB. These dense similarity matrices are temporary and are discarded after graph construction. In contrast, sparse graph storage increases only from 8.03MB to 33.12MB, approximately following the item-count growth because each item retains only a fixed number of top-k neighbors.

Overall, this analysis empirically confirms that SIGNAL contains quadratic preprocessing operations with respect to item count. Nevertheless, these costs are incurred only before training, and the resulting sparse graphs are reused without storing or recomputing dense item–item similarity matrices during repeated training.

4.7. Sparsity and noise conditions analysis (RQ5)

Sparsity Conditions. To explore recommendation capability of SIGNAL under sparse scenarios, we partition the Baby and Sports datasets into discrete intervals based on user interaction counts: (0,5], (5,10], (10,20], and (>20), and conduct comparative experiments with state-of-the-art MRs including NEGCL (Shi et al., 2025), LGMRec (Guo et al., 2024), and FREEDOM (Zhou & Shen, 2023). As shown in Fig. 11, where the bars indicate the number of users in each sparsity interval, for example, the first bars contains users interacting with 0–5 items, and the lines represent model performance. Notably, SIGNAL's performance curve remains consistently above all SOTA MRs in every sparsity levels. Its performance advantage is most pronounced in the most challenging extreme sparsity interval (0,5], where user interactions are scarcest. This result is critical, as it demonstrates that SIGNAL provides more accurate recommendations for cold-start users by effectively mitigating semantic ambiguity caused by sparse interactions. This capability stems from its holistic-to-partial granularity semantic alignment of modalities, which enhances the coherence of ID and multimodal embeddings of items and maximizes shared semantics to mitigate data sparsity.

Noise Conditions. To explore the noise resilience of SIGNAL, we inject random Gaussian noise into the visual and textual features of the Baby dataset, simulating different noise scenarios by varying the injection ratios (0.05, 0.1, 0.15, 0.2), and conduct comparative experiments with SOTA MRs including NEGCL (Shi et al., 2025), LGMRec (Guo et al., 2024), and FREEDOM Zhou and Shen (2023). Fig. 12 demonstrates SIGNAL's consistent superiority over SOTA baselines across noise conditions. Specifically, SIGNAL maintains the most stable performance curve under both visual and textual noise. While all models exhibit greater sensitivity to textual noise, SIGNAL's performance drop remains the smallest, particularly outperforming FREEDOM which suffers the most severe degradation due to its heavy reliance on the textual modality during graph fusion. These results underscore the critical role of SIGNAL's partial-granularity alignment strategy. By resolving nuanced inter-modal conflicts through its lightweight attentional diffusion module, this strategy reinforces semantic consistency among modalities, reduces noise susceptibility, and proves crucial when textual features are compromised. Notably, performance degradation is less severe with visual noise than with textual noise across all methods, suggesting greater reliance on textual information for item representation enhancement.

4.8. Case study of partial semantic conflict mitigation

To intuitively illustrate how LAD mitigates nuanced cross-modal conflicts, we conduct a case study on Item 21,257 on Clothing dataset, as shown in Fig. 13. At the content level, the visual and textual modalities exhibit consistent holistic semantics: both describe a women's red full-length dress. However, a clear partial inconsistency exists in the local pattern attribute. The product image presents a horizontal *Striped Pattern*, whereas the textual description specifies *Floral Print*. When such inconsistent cues are fused into a shared item representation, they can introduce semantic interference and obscure the represented item attributes.

At the representation level, the heatmaps visualize the element-wise absolute differences between the visual and textual representations before and after LAD. Although individual latent dimensions cannot be directly mapped to specific attributes such as *striped* or *floral*, the overall weakening of the heatmap values indicates that the visual–textual representation discrepancy is reduced after refinement. It should be emphasized that LAD does not directly minimize these element-wise differences or assume that all cross-modal discrepancies are harmful; the heatmaps are used only as a diagnostic visualization of the conflict-refinement effect. Excessively suppressing cross-modal discrepancies could homogenize modality-specific representations and lead to over-smoothing. Therefore, LAD retains a residual connection to the original modality-specific representations to preserve their complementary semantics, while its effectiveness is ultimately evaluated by the resulting recommendation performance. To isolate the effect of conflict

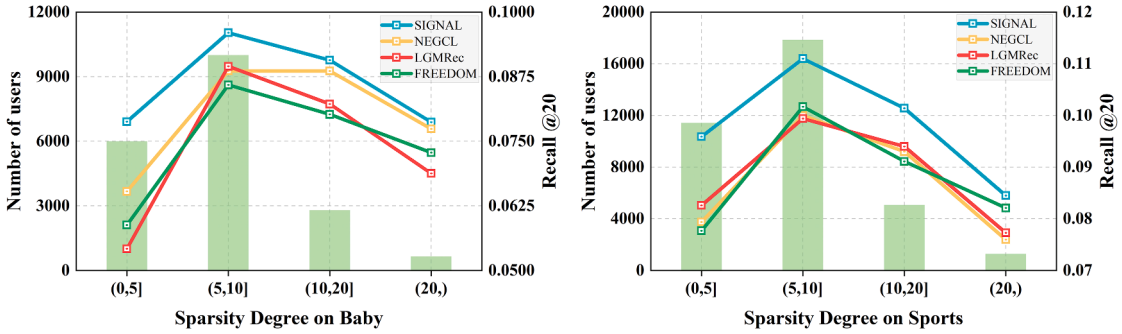


Fig. 11. Sparsity degree analysis on Baby and Sports.

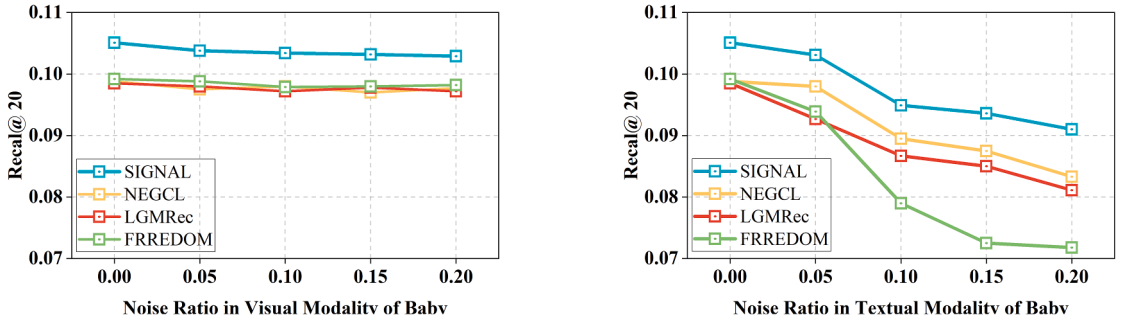


Fig. 12. Noise degree analysis in visual and textual modality of the Baby.

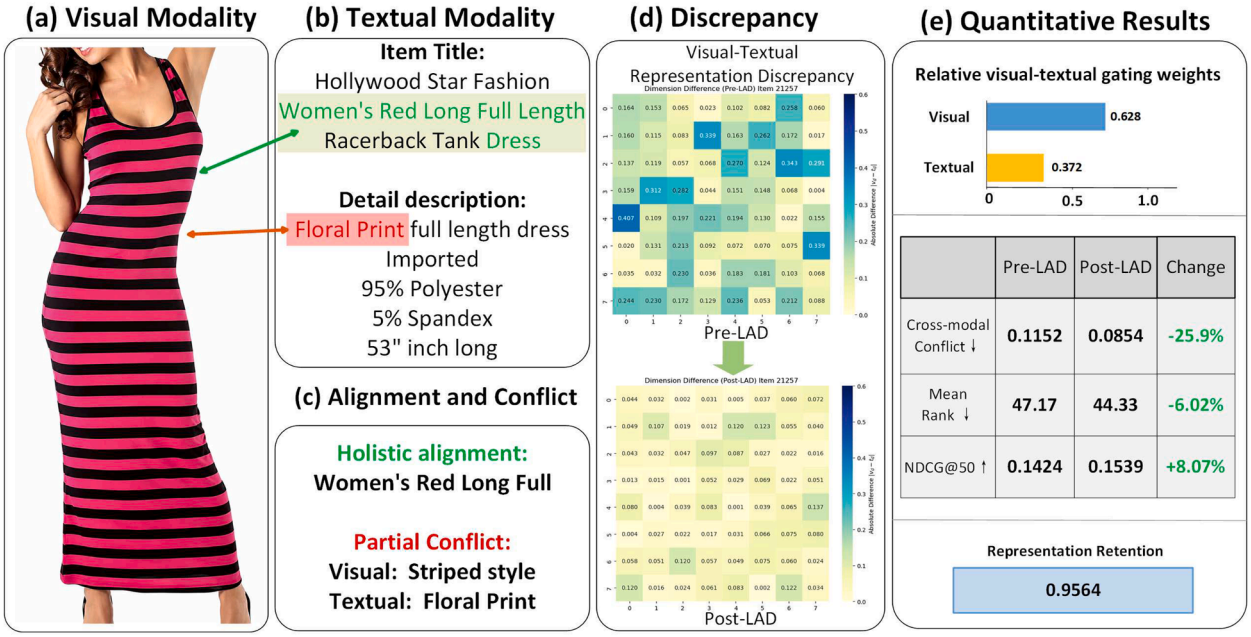


Fig. 13. Case study of partial semantic conflict mitigation for Item 21,257 on Clothing dataset. The heatmaps show the element-wise absolute differences between the visual and textual representations before and after LAD. The quantitative results are calculated over the six users interacting with this item.

refinement, the heatmaps visualize the restored representations before the residual connection is applied.

For this case, we further report relative visual–textual gating weights to illustrate the reliability-aware guidance during restoration. The visual coefficient is larger than the textual coefficient (0.628 vs. 0.372), indicating that the restoration process places greater emphasis on visual semantics when reconciling the conflict between the visually observed *Striped Pattern* and the textual attribute *Floral Print*.

We further quantify the overall cross-modal conflict of item i as the average pairwise cosine distance among its visual, textual, and behavior-aware (Co-occurrence) representations:

$$C_i = \frac{1}{3}\left[1-\cos(\mathbf{h}_i^v,\mathbf{h}_i^t)+1-\cos(\mathbf{h}_i^v,\mathbf{h}_i^c)+1-\cos(\mathbf{h}_i^t,\mathbf{h}_i^c)\right], \tag{26}$$

where a smaller C_i indicates stronger cross-modal consistency. For Item 21257, C_i decreases from 0.1152 to 0.0854, corresponding to a reduction of 25.87% after LAD. Meanwhile, the representation-retention score

reaches 0.9564, indicating that LAD mitigates conflict-induced interference while preserving most of the original modality-specific information.

We further evaluate the ranking performance of Item 21,257 over the six users who interacted with it. For each user, we record the position of Item 21,257 in the predicted ranking over all candidate items and average these positions as the Mean Rank, where a lower value indicates a better ranking. After LAD, the Mean Rank decreases from 47.17 to 44.33, indicating that the item is promoted by 2.84 positions on average, corresponding to a relative improvement of 6.02%. NDCG@50 further increases from 0.1424 to 0.1539, corresponding to a relative improvement of 8.07%. Together with the improved Mean Rank, this result suggests that mitigating partial cross-modal conflicts yields a more accurate and less ambiguous representation of the item's semantics, facilitating more reliable matching between the item and user preferences. Although this item-level case involves only six users, it provides an intuitive example complementing the overall experimental results.

5. Conclusion

In this paper, we propose SIGNAL, a multi-granularity semantic alignment framework for multimodal recommendation. SIGNAL addresses two limitations in existing multimodal semantic learning: the weak coupling between graph-based encoding and contrastive alignment, and the persistence of nuanced cross-modal conflicts after holistic alignment. For holistic semantic learning, PGE adaptively fuses modality-specific graph structures according to cross-modal neighborhood consistency, producing preliminarily aligned representations during encoding. HSC subsequently derives hierarchical item relations from the semantic neighborhoods established during graph encoding and performs differentiated contrast to preserve these neighborhoods during alignment. For partial semantic learning, LAD models residual cross-modal conflicts as semantic interference and restores the representations under reliability-aware guidance. By leveraging the holistically aligned representations as a favorable initialization, LAD requires only a few restoration steps and maintains moderate computational overhead. Extensive experiments on four benchmark datasets demonstrate the effectiveness of SIGNAL, while ablation, robustness, visualization, compatibility, and efficiency analyses further validate the roles and interactions of the proposed mechanisms.

Several limitations remain. First, PGE involves pairwise item similarity computation and top-k graph construction, resulting in quadratic preprocessing complexity with respect to the number of items. Although these graphs are constructed only once, stored sparsely, and reused throughout training, the temporary dense preprocessing cost can become substantial on ultra-large-scale datasets. Due to current hardware limitations, we have not yet validated SIGNAL in such settings. Second, LAD introduces an additional restoration network and several hyperparameters, and the magnitude of its improvement varies across datasets. Its benefit is more pronounced when richer multimodal signals leave greater room for partial conflict refinement, while the gains can be moderate when most semantic discrepancies have already been resolved by graph encoding and holistic alignment. Future work will investigate approximate similarity search and scalable sparse graph construction, together with adaptive restoration that selects diffusion steps according to conflict intensity and reduces manual hyperparameter tuning.

CRediT authorship contribution statement

Meizi Li: Conceptualization, Writing – review and editing, Funding acquisition; **Hang Gao:** Methodology, Investigation, Visualization, Writing – original draft; **Bo Zhang:** Conceptualization, Funding acquisition; **Yang Yu:** Formal analysis, Investigation, Validation; **Guojian Zou:** Formal analysis, Validation, Supervision; **Ling Ding:** Conceptualization, Methodology, Validation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Bansal, A., Borgnia, E., Chu, H.-M., Li, J., Kazemi, H., Huang, F., Goldblum, M., Geiping, J., & Goldstein, T. (2023). Cold diffusion: Inverting arbitrary image transforms without noise. *Advances in Neural Information Processing Systems*, *36*, 41259–41282.

Chen, J., Zhang, H., He, X., Nie, L., Liu, W., & Chua, T.-S. (2017). Attentive collaborative filtering: Multimedia recommendation with item-and component-level attention. In *Proceedings of the 40th International ACM SIGIR conference on research and development in information retrieval* (pp. 335–344).

Cheng, R., & Song, B. (2025). CDMR2F: Correlation-guided denoising multimodal robust fusion framework for multimodal recommendation. *Expert Systems with Applications*, 130530.

Cheng, X., Zhong, Y., Dai, Y., Ji, P., & Li, H. (2019). Noise-aware unsupervised deep lidar-stereo fusion. In *Proceedings of the IEEE/CVF Conference on computer vision and pattern recognition* (pp. 6339–6348).

Guo, Z., Li, J., Li, G., Wang, C., Shi, S., & Ruan, B. (2024). LGMREC: Local and global graph learning for multimodal recommendation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 8454–8462). (*vol. 38*).

He, Q., Li, K., Zhang, Y., Li, X., & Zhou, W. (2026). Smooth diffusion model for multimodal recommendation. *Knowledge-Based Systems*, *331*, 114807. https://doi.org/10.1016/j.knosys.2025.114807

He, R., & McAuley, J. (2016). VBPR: Visual bayesian personalized ranking from implicit feedback. In *Proceedings of the AAAI conference on artificial intelligence*. (*vol. 30*).

He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., & Wang, M. (2020). LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in information retrieval* (pp. 639–648).

Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, *33*, 6840–6851.

Jiang, Y., Xia, L., Wei, W., Luo, D., Lin, K., & Huang, C. (2024). Diffmm: Multi-modal diffusion model for recommendation. In *Proceedings of the 32nd ACM International conference on multimedia* (pp. 7591–7599).

Kim, Y., Jo, D., Jeon, H., Kim, T., Ahn, D., Kim, H. et al. (2023). Leveraging early-stage robustness in diffusion models for efficient and high-quality image synthesis. *Advances in Neural Information Processing Systems*, *36*, 1229–1244.

Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. arXiv:1412.6980

Li, C., Zhou, W., Zhang, Y., Hu, J., Shen, H., & Wen, J. (2026a). MSCF-Net: Multi-scale frequency denoising and co-frequency enhancement network for multimodal recommendation. *Expert Systems with Applications*, 131159.

Li, J., Wang, S., Zhang, Q., Yu, S., & Chen, F. (2025a). Generating with fairness: A modality-diffused counterfactual framework for incomplete multimodal recommendations. In *Proceedings of the ACM on web conference 2025* (pp. 2787–2798).

Li, K., Zhang, Y., He, Q., & Li, X. (2026b). Leveraging hierarchy-aware diffusion model and knowledge-enhanced contrastive learning for recommendation. *Knowledge and Information Systems*, *68* (1), 34. https://doi.org/10.1007/s10115-025-02639-4

Li, K., Zhang, Y., Li, X., Yuan, M., & Zhou, W. (2025b). Mask diffusion-based contrastive learning for knowledge-aware recommendation. *IEEE Transactions on Knowledge and Data Engineering*, *37* (9), 5407–5419. https://doi.org/10.1109/TKDE.2025.3582767

Li, K., Zou, J., Lin, Q., Guo, W., He, Q., & Yang, Y. (2026c). ScotRec: Social chain-of-thought LLM reasoning for recommendation. In *Proceedings of the ACM web conference 2026* (pp. 6242–6251). https://doi.org/10.1145/3774904.3792313

Li, X., Thickstun, J., Gulrajani, I., Liang, P. S., & Hashimoto, T. B. (2022). Diffusion-lm improves controllable text generation. *Advances in Neural Information Processing Systems*, *35*, 4328–4343.

Liu, F., Cheng, Z., Sun, C., Wang, Y., Nie, L., & Kankanhalli, M. (2019). User diverse preference modeling by multimodal attentive metric learning. In *Proceedings of the 27th ACM International conference on multimedia* (pp. 1526–1534).

Liu, J., Zhang, Y., He, Q., Yang, R., Wang, X., & Zhou, W. (2025). Prototype learning based hierarchical decoupling for multimodal recommendation. *Expert Systems with Applications*, 130763.

Liu, Q., Hu, J., Xiao, Y., Zhao, X., Gao, J., Wang, W., Li, Q., & Tang, J. (2024). Multimodal recommender systems: A survey. *ACM Computing Surveys*, *57* (2), 1–17.

Mathieu, M., Couprie, C., & LeCun, Y. (2015). Deep multi-scale video prediction beyond mean square error. arXiv:1511.05440

Qi, Y., Zhang, Q., Lin, X., Su, X., Zhu, J., Wang, J., & Li, J. (2025). Seeing beyond noise: Joint graph structure evaluation and denoising for multimodal recommendation. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 12461–12469). (*vol. 39*).

Qi, Y., Zhang, Q., Ma, X., Feng, X., Li, Q., & Huang, Y. (2026). Cross-modal local interest contrast with dual-graph denoising for multimodal recommendation. *Signal, Image and Video Processing*, *20* (1), 42.

Rendle, S., Freudenthaler, C., Gantner, Z., & Schmidt-Thieme, L. (2009). BPR: Bayesian personalized ranking from implicit feedback. In *Proceedings of the twenty-fifth conference on uncertainty in artificial intelligence* (pp. 452–461).

Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on computer vision and pattern recognition* (pp. 10684–10695).

Shi, K., Zhang, Y., Zhang, M., Xiao, K., Hou, X., & Li, Z. (2025). Noise-enhanced graph contrastive learning for multimodal recommendation systems. *Knowledge-Based Systems*, 113766.

Song, J., Meng, C., & Ermon, S. (2021). Denoising diffusion implicit models. In *International conference on learning representations (ICLR)*.

Song, Q., Hu, J., Xiao, L., Sun, B., Gao, X., & Li, S. (2025). DiffCL: A diffusion-based contrastive learning framework with semantic alignment for multimodal recommendations. *IEEE Transactions on Neural Networks and Learning Systems*, *36* (10), 18587–18597.

Tao, Z., Liu, X., Xia, Y., Wang, X., Yang, L., Huang, X., & Chua, T.-S. (2022). Self-supervised learning for multimedia recommendation. *IEEE Transactions on Multimedia*, *25*, 5107–5116.

van den, O. A., Li, Y., & Vinyals, O. (2018). Representation learning with contrastive predictive coding. arXiv:1807.03748

van der, M. L., & Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, *9* , 2579–2605.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Advances in neural information processing systems*, *vol. 30*.

Wei, W., Huang, C., Xia, L., & Zhang, C. (2023). Multi-modal self-supervised learning for recommendation. In *Proceedings of the ACM web conference 2023* (pp. 790–800).

Wei, Y., Wang, X., Nie, L., He, X., & Chua, T.-S. (2020). Graph-refined convolutional network for multimedia recommendation with implicit feedback. In *Proceedings of the 28th ACM International conference on multimedia* (pp. 3541–3549).

Wei, Y., Wang, X., Nie, L., He, X., Hong, R., & Chua, T.-S. (2019). MMGCN: Multi-modal graph convolution network for personalized recommendation of micro-video. In *Proceedings of the 27th ACM International conference on multimedia* (pp. 1437–1445).

Wei, Y., Xu, Y., Zhu, L., Ma, J., & Huang, J. (2024). Fummer: A fine-grained self-supervised momentum distillation framework for multimodal recommendation. *Information Processing & Management*, *61* (5), 103776.

Xv, G., Li, X., Xie, R., Lin, C., Liu, C., Xia, F., Kang, Z., & Lin, L. (2024). Improving multi-modal recommender systems by denoising and aligning multi-modal content and user feedback. In *Proceedings of the 30th ACM SIGKDD Conference on knowledge discovery and data mining* (pp. 3645–3656).

Yu, P., Tan, Z., Lu, G., & Bao, B.-K. (2023). Multi-view graph convolutional network for multimedia recommendation. In *Proceedings of the 31st ACM International conference on multimedia* (pp. 6576–6585).

Zhang, J., Zhu, Y., Liu, Q., Wu, S., Wang, S., & Wang, L. (2021). Mining latent structures for multimedia recommendation. In *Proceedings of the 29th ACM International conference on multimedia* (pp. 3872–3880).

Zhang, J., Zhu, Y., Liu, Q., Zhang, M., Wu, S., & Wang, L. (2022). Latent structure mining with contrastive modality fusion for multimedia recommendation. *IEEE Transactions on Knowledge and Data Engineering*, *35* (9), 9154–9167.

Zhang, K., Zuo, W., Chen, Y., Meng, D., & Zhang, L. (2017). Beyond a gaussian denoiser: Residual learning of deep cnn for image denoising. *IEEE Transactions on Image Processing*, *26* (7), 3142–3155.

Zhang, Q., Wei, Y., Han, Z., Fu, H., Peng, X., Deng, C., Hu, Q., Xu, C., Wen, J., Hu, D. et al. (2024). Multimodal fusion on low-quality data: A comprehensive survey. arXiv:2404.18947

Zhou, H., Zhou, X., Zeng, Z., Zhang, L., & Shen, Z. (2023a). A comprehensive survey on multimodal recommender systems: Taxonomy, evaluation, and future directions. arXiv:2302.04473

Zhou, H., Zhou, X., Zhang, L., & Shen, Z. (2023b). Enhancing dyadic relations with homogeneous graphs for multimodal recommendation. In *ECAI 2023* (pp. 3123–3130).

Zhou, X., & Shen, Z. (2023). A tale of two graphs: Freezing and denoising graph structures for multimodal recommendation. In *Proceedings of the 31st ACM International conference on multimedia* (pp. 935–943).

Zhou, X., Zhou, H., Liu, Y., Zeng, Z., Miao, C., Wang, P., You, Y., & Jiang, F. (2023c). Bootstrap latent representations for multi-modal recommendation. In *Proceedings of the ACM web conference 2023* (pp. 845–854).